

EMI & EMC

Design Guidelines

for PCBs



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About the author



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Karen Burnham, President and Chief EMC Engineer at EMC United, has worked in the aerospace, defense, automotive, and consulting industries since 1996. Ms. Burnham holds a BS in Physics and an MS in Electrical Engineering. She is widely known for her ability to translate complex EMC concepts into practical, plain-English guidance for hardware designers.

Her experience includes managing EMC requirements and test planning for NASA programs, including the Dream Chaser spacecraft. She has done troubleshooting on electric vehicles for Ford Motor Company and other OEMs. Karen has initiated innovative SBIRs and STTRs through government centers, contributed to classified projects, and consulted across a broad range of industries (space systems to commercial electronics).

She is deeply involved in standards development and currently serves as the Vice President of Standards for the IEEE EMC Society, participating on multiple international standards committees.

About EMC United

Ms. Burnham founded EMC United with the aim of making electromagnetic compatibility less painful for engineers, managers, and technicians. Through education, consulting, and troubleshooting, EMCU focuses on helping organizations resolve EMC challenges as quickly and efficiently as possible. The company's approach emphasizes practical understanding and early problem-solving, enabling teams to approach EM compliance with greater confidence. EMC United is a woman-owned small business based near Denver, Colorado.

For more, visit www.emcunited.com

Table of Contents

01	What is EMI?	05
	1.1 How EMI spreads	05
	1.2 How to identify EMC requirements for your project	08
	1.3 What EMC tests simulate	08
	1.4 Common EMC Standards	08
02	How to design for electromagnetic resilience	12
	2.1 Identify and protect critical functions early	12
	2.2 Implement software techniques for EMI resilience	13
03	What are the sources of electromagnetic interference?	16
	3.1 Switching operations	16
	3.2 Pulse-width modulation (PWM)	21
	3.3 Motors	22
04	What structures cause radiated EMI in PCBs?	24
	4.1 Cables	24
	4.2 Ribbon cables	25
	4.3 Traces	29
	4.4 Ungrounded components	29
	4.5 Loops	30
	4.6 Slots and apertures	31
	4.7 Electrical bonding	35

05

Design guidelines to ensure EMC in your PCBs

5.1 Use transmission lines for field containment

5.2 Route traces to minimize radiation and coupling

5.3 Design stack-ups with solid reference planes

5.4 Add stitching vias near layer transitions

5.5 Use filters to control conducted EMI

5.6 Implement shielding to confine EM fields

Conclusion

.....37

.....38

.....45

.....47

.....51

.....54

.....59

.....65

1. What is EMI?

Electromagnetic (EM) interference happens when EM energy from one device disrupts another device's operation. This energy can travel through the air (radiation) or through wires and cables (conduction).

EMI problems most commonly occur between 9 kHz and 6 GHz. This guide focuses on that frequency range.

Above 6 GHz, interference usually stays within a single PCBA. These higher-frequency problems are usually considered signal integrity issues rather than EMI, which more commonly deals with interactions between different systems or different parts of a larger system.

Common sources of EMI in this frequency range include:

- AM and FM radio stations
- Emergency services radios
- WiFi and Bluetooth
- Cellular phones
- Switched-mode power supplies (SMPS)
- DC-to-DC converters
- Motors and inverters

Real-world examples

You may have experienced EMI in daily life:

- A microwave oven disrupting your home WiFi.
- Airplane crew asking passengers to turn off phones during takeoff and landing to prevent interference with safety-critical aircraft systems.

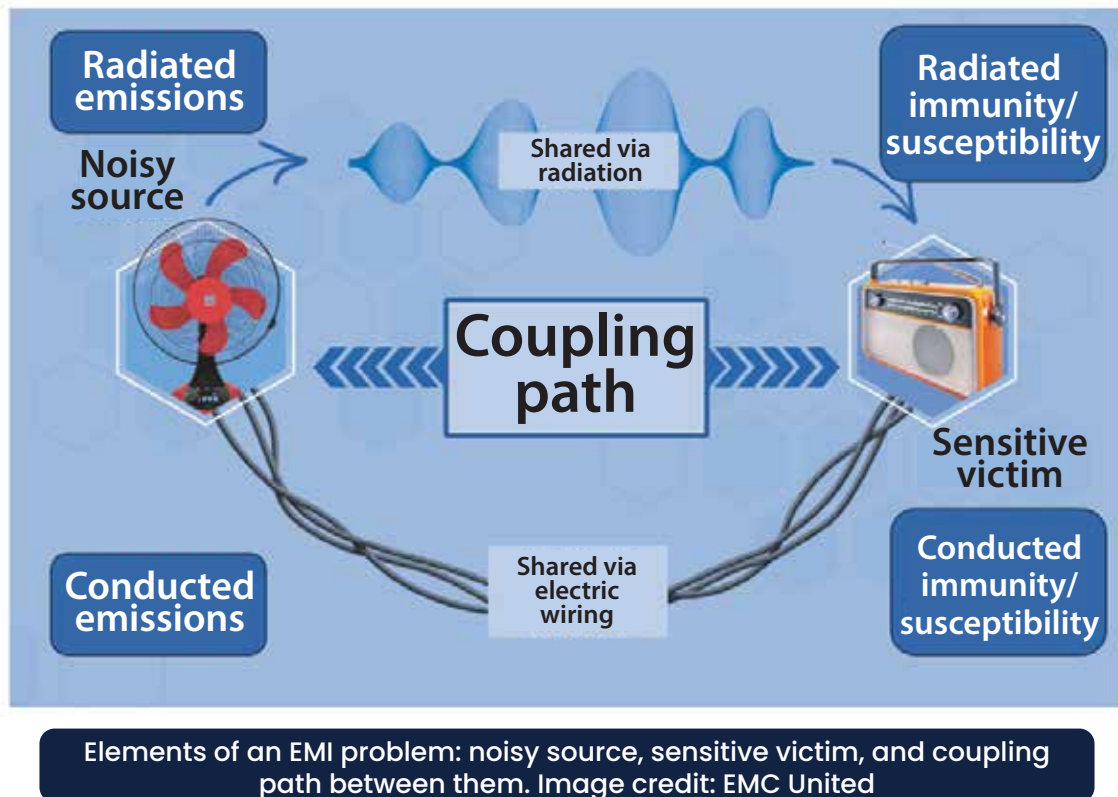
1.1 How EMI spreads

EMI occurs when electromagnetic fields interact with electronic components in unwanted ways. Fields can be created on a board and then:

- Radiate directly into the air (radiated emissions).
- Couple into wires and affect other components (conducted emissions).

External EM fields can also enter your system through:

- The air, affecting circuit board traces or wiring directly (radiated immunity or radiated susceptibility).
- Shared power lines or cables (conducted immunity or conducted susceptibility).



The fundamental challenge

All EMI problems involve controlling electromagnetic fields. They must be contained and directed properly. A field's propagation and its effect on systems change with every design choice:

- Requirements definition (EMC targets and operating conditions)
- Mechanical enclosure design and installation
- Software implementation
- PCB layout
- Cable routing

Using conductive materials (metal planes, traces, shields, enclosures) to control where fields can travel is one key to EMI reduction. Defining requirements correctly and implementing robust software are also crucial.

Can we eliminate EMI completely?

No, EMI cannot be eliminated entirely. However, we can design electronics to:

- Be less vulnerable to EMI from other sources.
- Generate less interference that affects other devices.

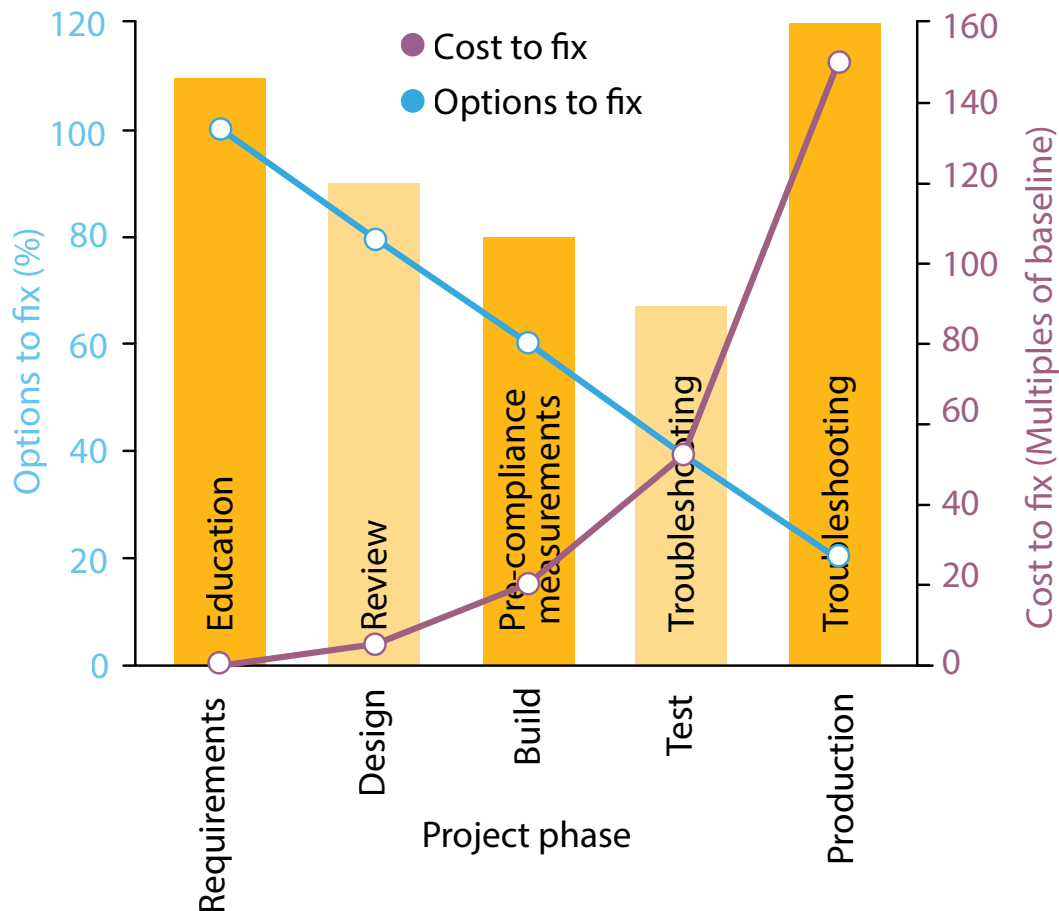
This is called achieving electromagnetic compatibility (EMC).

Why EMC matters for your project

Most units fail the EMC test on their first attempt. Industry data shows that 60–90% of products do not pass the compliance test. This leads to:

- Project delays
- Budget overruns
- Expensive redesigns with limited options

EMI issues are much easier and less expensive to fix early in the project. As development progresses, design flexibility decreases, and the cost of fixes rises.



Consequences of failing to consider EMC early. Image credit: EMC United

Following this guide will significantly improve your chances of passing EMC tests on the first try.

REMEMBER: EMC problems are difficult and expensive to fix late in the project.

1.2 How to identify EMC requirements for your project

Electromagnetic compatibility requirements depend on three factors:

1. What type of product are you building?
2. How will the product be used?
3. Where will the product be sold?

Different countries and regions have different regulations. This means the same product may need different tests depending on where it will be sold.

1.3 What EMC tests simulate

EMC tests recreate various electromagnetic disturbances that your product might encounter, such as:

- Magnetic fields from electrical wires
- Radiated and conducted EM noise
- Voltage drops from power interruptions
- Fast transients from electrical switches
- Electromagnetic surges from lightning strikes
- Electrostatic discharges (ESD) from static electricity

These tests are designed to verify that a product can operate reliably in its intended environment without causing or suffering from interference.

Each regulatory body has its own set of standards. Finding the correct requirements can be confusing. For this reason, applicable standards should be identified early in the project.

We strongly recommend: Do your own research or hire an EMC expert or test lab early in your project. Identify the required standards before you build prototypes for formal testing. This will save time and money.

1.4 Common EMC Standards

This section covers three commonly used EM compatibility benchmarks. Understanding these will help you identify which regulations apply to your product and what testing you'll need to prepare for.

1. FCC Part 15: Radio-frequency devices

It is a section of the Federal Communications Commission (FCC) regulations that governs the operation of consumer electronic devices in the United States. Very similar emission limits and test methods for global markets are found in CISPR 32.

FCC Part 15 sets standards for:

- Devices that emit radio frequency energy (including intentional, unintentional, and incidental radiators). It ensures they do not cause harmful interference to licensed radio services and must accept any interference received.
- Applications that generate frequencies of 9 kHz or higher. The standard makes it mandatory to test these devices for FCC compliance, even if they qualify for other test exemptions.

Failure to comply can result in fines, product recalls, and halting distribution.

Device classifications as per FCC Part 15 include:

- **Intentional radiators:** Devices that intentionally emit radio frequency energy, such as WiFi routers and Bluetooth devices.
- **Unintentional radiators:** Applications that generate radio frequency energy as a byproduct, such as computers and TV receivers.
- **Incidental radiators:** Systems that emit radio frequency energy unintentionally, such as electric motors and light dimmers.

Test methods for verifying compliance with FCC Part 15 can be found in ANSI C63.4.

2. CISPR 11: Industrial, scientific, and medical (ISM) equipment

CISPR refers to the International Special Committee on Radio Interference, established in 1933. It develops standard measurement methods for radio frequency emissions.

CISPR 11 is an EMC standard specifically for industrial, scientific, and medical (ISM) equipment operating from 9 kHz to 400 GHz. It is the primary EMC standard referenced by IEC 60601 for medical devices.

The standard specifies:

- Conducted and radiated emission limits for each equipment class, with different limits depending on the measurement distance and test chamber used.
- Requirements for test equipment, ambient noise levels, measurement instruments, and test setups for various equipment configurations.

CISPR organizes equipment by type and environment:

- **Equipment classification:**
 - Group 1: Medical electrical equipment, machine tools, and scientific equipment.
 - Group 2: ISM RF-generating equipment like microwave ovens, inductive charging equipment, and electric welding equipment.
- **Equipment classes:**
 - Class A: Equipment suitable for industrial and heavy commercial environments.
 - Class B: Devices intended for residential and light commercial environments, with stricter emission limits than Class A.

3. MIL-STD-461: Military EMC testing

It is a standard for EMC testing maintained by the U.S. Department of Defense to ensure military equipment does not emit EMI that could disrupt other devices and is not susceptible to EMI from other sources.

MIL-STD-461 certification is essential for mission-critical systems to avoid disruptions, mission failures, and data loss due to EMI. It is widely used across the defense and aerospace industries.

This standard defines 19 EMC test methods, grouped into four categories:

- Conducted emissions (CE)
- Radiated emissions (RE)
- Conducted susceptibility (CS)
- Radiated susceptibility (RS)

Only a subset of these tests applies to any given piece of equipment, depending on its function and installation environment.

Table IV from MIL-STD-461F outlines the applicable emission and susceptibility test methods.

Test category	What does the test evaluate?
CE101	Conducted emissions, audio frequency currents, power leads
CE102	Conducted emissions, radio frequency potentials, power leads
CE106	Conducted emissions, antenna port

Test category	What does the test evaluate?
CS101	Conducted susceptibility, power leads
CS103	Conducted susceptibility, antenna port, intermodulation
CS104	Conducted susceptibility, antenna port, rejection of undesired signals
CS105	Conducted susceptibility, antenna port, cross-modulation
CS109	Conducted susceptibility, structure current
CS114	Conducted susceptibility, bulk cable injection
CS115	Conducted susceptibility, bulk cable injection, impulse excitation
CS116	Conducted susceptibility, damped sinusoidal transients, cables, and power leads
CS117	Conducted susceptibility, lightning-induced transients, cables, and power leads
CS118	Conducted susceptibility, personnel borne ESD
RE101	Radiated emissions, magnetic field
RE102	Radiated Emissions, electric field
RE103	Radiated emissions, antenna spurious and harmonic outputs
RS101	Radiated susceptibility, magnetic field
RS103	Radiated susceptibility, electric field
RS105	Radiated susceptibility, transient electromagnetic field

Proper application of these tests ensures that military and aerospace equipment meets stringent EM compatibility requirements, preventing interference and ensuring reliable operation in various environments.

Important note on tailoring: These requirements are generic, since the types of equipment that could fall under MIL-STD-461 range from a walkie-talkie to an aircraft carrier. Each program must tailor the applicable tests, setups, and limits to its specific hardware.

Remember: If your product will be sold in multiple countries, you may need to comply with multiple standards. For example, a device sold in both the United States and Europe might need to comply with both FCC Part 15 and CISPR 32.

2. How to design for electromagnetic resilience

Designing for EM resilience means ensuring a product remains safe, functional, and recoverable when disturbances occur.

EMC testing is important, but it cannot cover every possible real-world situation. Tests use specific frequencies, field strengths, and conditions. Your product will encounter many situations that fall outside typical testing scenarios.

We need to ensure that the design is resilient to electromagnetic effects instead of designing only to pass a test.

The following design factors provide a structured approach to building EMI-resilient systems:

2.1 Identify and protect critical functions early

For every project, identify the functions that are critical for:

- Safety
- Proper operation of the device

For each function, design the system so that when EMI occurs, the unit will remain safe and operational.

IEEE 1848 provides comprehensive guidance on this topic. The full title is IEEE Standard for Techniques and Measures to Manage Functional Safety and Other Risks with Regard to Electromagnetic Disturbances.

This standard offers methods to ensure safety even when EMI is present.

Key design considerations include:

2.1.1 Understand the operating environment

Start by considering where and how the device will be used:

- Will it be exposed to lightning?
- Will users touch it and generate electrostatic discharge (ESD)?
- Will it operate near military jamming equipment?
- Will it be near electric vehicle motors?
- Are there other unusual electromagnetic conditions in the environment?

Design the system with the above environmental factors in mind.

2.1.2 Account for the device lifecycle

EMC performance can degrade over time. Consider how aging affects the system:

- Fasteners and connectors loosen with vibration and use.
- Corrosion develops, especially in humid or marine environments.
- Components gradually degrade or fail.

Ask yourself: Will the design still provide adequate EMC performance as the hardware ages?

2.1.3 Use redundancy and error detection

If a signal or input becomes corrupted by EMI, can the system detect this problem?

Consider these approaches:

- Use multiple sensors or signals that are routed differently.
- Use different signal types that won't be affected by the same interference.
- Implement checking mechanisms to verify data integrity.

Example: If one temperature sensor gives an impossible reading, the system should recognize this and use a backup sensor or enter a safe mode.

2.1.4 Incorporate filtering and shielding

Filtering and shielding are most effective when planned early.

Key principles:

- Apply filtering at all power and I/O entry points.
- Use shielding to contain noisy circuits or protect sensitive ones.
- Ensure low-impedance grounding and continuity for all shields.

Late-stage additions often create new problems, such as resonance, ground loops, or unintended antennas.

This is discussed in detail in sections 5.5 and 5.6.

2.2 Implement software techniques for EMI resilience

Hardware alone is not sufficient. Software plays a critical role in detecting EMI-related faults and recovering from them quickly.

Consider these approaches:

2.2.1 Set time limits

Software should never wait indefinitely for an event to occur. Always set a maximum time limit.

Example: If software is waiting for a sensor response:

- Bad approach: Wait indefinitely.
- Good approach: Wait for 100 milliseconds maximum. If no response, stop waiting and try an alternative action.

If a wait time exceeds its limit, the software should stop waiting and try something else.

2.2.2 Use a watchdog timer

A watchdog timer is a safety feature that automatically restarts the system if the software stops working correctly.

Think of the watchdog as a guard that repeatedly asks, "Is the program still running?" Every few seconds, the main program must respond to the watchdog. If the program doesn't respond (because it crashed or got stuck), the watchdog automatically resets the entire system.

This prevents the system from being permanently stuck in a bad state due to EMI.

2.2.3 Add error-checking to data

Before saving important information to memory, add extra bits that help detect corruption.

How it works:

- Before writing data, calculate a checksum or error-correcting code.
- Store both the data and the error-checking bits.
- When reading the data later, recalculate the checksum.
- Compare the stored checksum with the newly calculated checksum.
- If they don't match, the data is corrupted, don't use it.

This prevents the system from using distorted data that could cause incorrect operation.

2.2.4 Capture and store input signals properly

Every input pin must have a defined state at all times.

Best practices:

- Read each input pin at a specific moment in time.
- Store the reading in memory immediately.
- Never leave input pins unconnected or undefined.
- Each input must always be either HIGH or LOW.

Floating inputs (pins not connected to anything) act like antennas. They can pick up electrical noise from the environment and produce unpredictable voltage levels. This causes random, unreliable behavior.

Use pull-up or pull-down resistors to ensure every input has a defined default state.

2.2.5 Consider using fiber optic

For particularly critical interfaces that must be completely immune to EMI, consider using fiber optic connections.

Why fiber optics

- Fiber optic cables carry signals as light, not electrical current.
- Light signals are inherently immune to EMI in these frequency ranges.
- Fiber optic cables completely contain their signal, with no leakage of electromagnetic fields.

Design philosophy. Assume EMI will occur. Design your system to detect it, survive it, and recover from it quickly. Don't assume passing EMC tests means your device will never encounter EMI problems in the field.

Note: These software techniques add resilience but are not substitutes for proper hardware design. Use both hardware and software methods together for the best results.

3. What are the sources of electromagnetic interference?

Certain parts of a design generate more high-frequency noise than others. Understanding where noise comes from helps you control it effectively.

Common noise sources include:

3.1 Switching operations

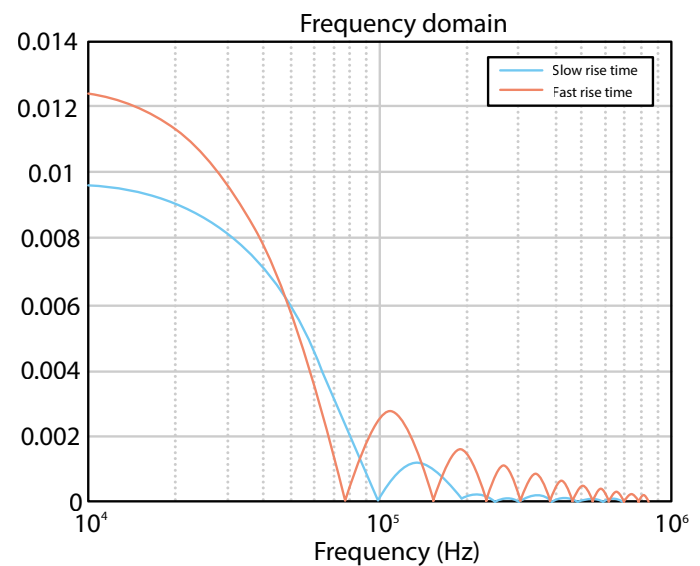
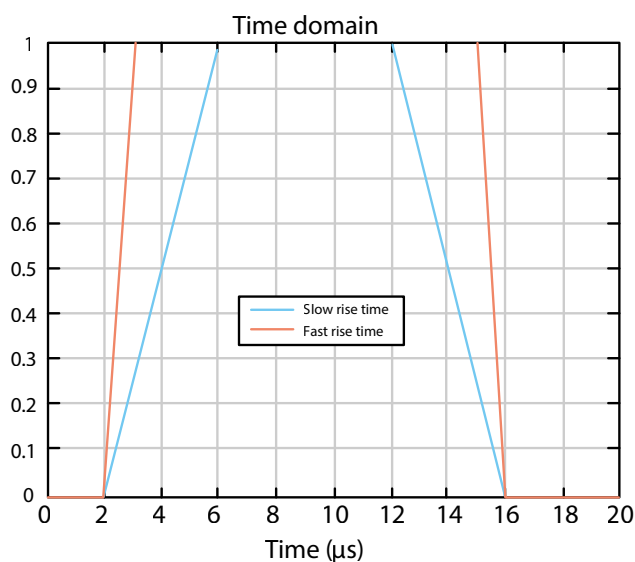
It is one of the most common causes of high-frequency noise. Switching happens whenever a signal changes its state from LOW to HIGH or HIGH to LOW. These operations occur in:

- Integrated circuits (ICs)
- Clock signals
- Switched-mode power supplies (SMPS)
- DC-to-DC converters

3.1.1 Why switching creates noise

Every switching operation generates high-frequency noise, even for a DC signal (like a clock or digital signal).

The faster the signal transitions between LOW and HIGH (i.e., faster rise and fall times), the more high-frequency noise is created.



Difference in high-frequency content between a signal with a faster and a slower rise time. Image credit: EMA.

Impact of new technologies

When you use faster components or new switching technologies, expect more high-frequency noise through:

- Faster, sharper clock operations
- Wide bandgap semiconductors (silicon carbide or gallium nitride)

Real-world example

Designs that previously passed EMC testing have failed badly after switching to silicon carbide (SiC) or gallium nitride (GaN) power devices.

These new semiconductors switch faster than older silicon devices by orders of magnitude, creating significantly more high-frequency noise.

Design recommendation

When possible, choose components with the slowest rise and fall times that still meet your requirements for:

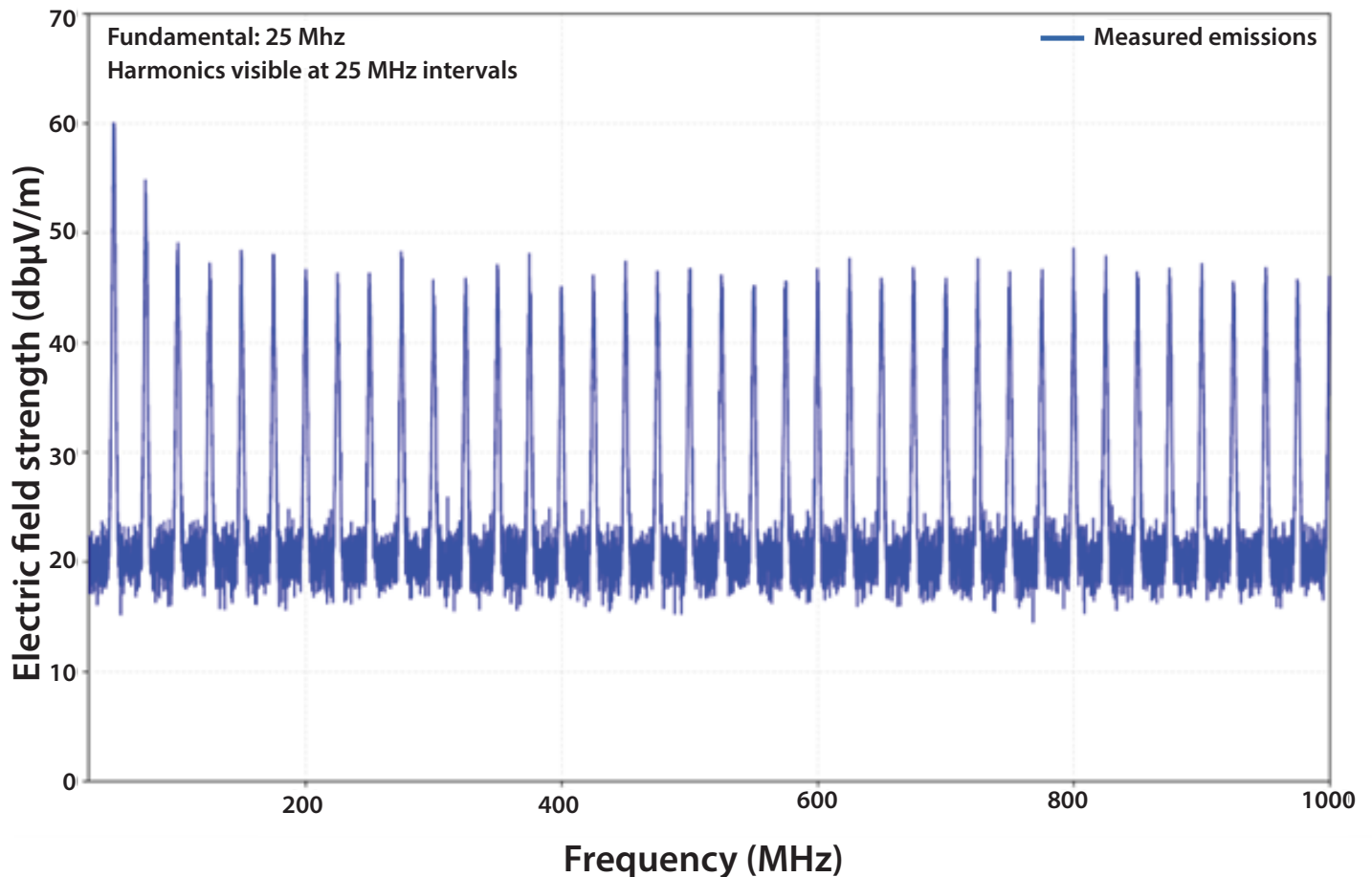
- Thermal performance
- Power efficiency
- Functional operation

Slower switching = less high-frequency noise = better EMC performance.

When very fast components are necessary, plan ahead for the high-frequency noise they will create. Include mitigation techniques in the design from the start.

What does switching noise look like?

Noise from switching operations typically looks like a series of sharp, evenly spaced peaks across a frequency plot.



Example radiated emissions plot from a 25 MHz clock signal.
Image credit: EMC United

3.1.2 How switching waveforms generate high-frequency noise

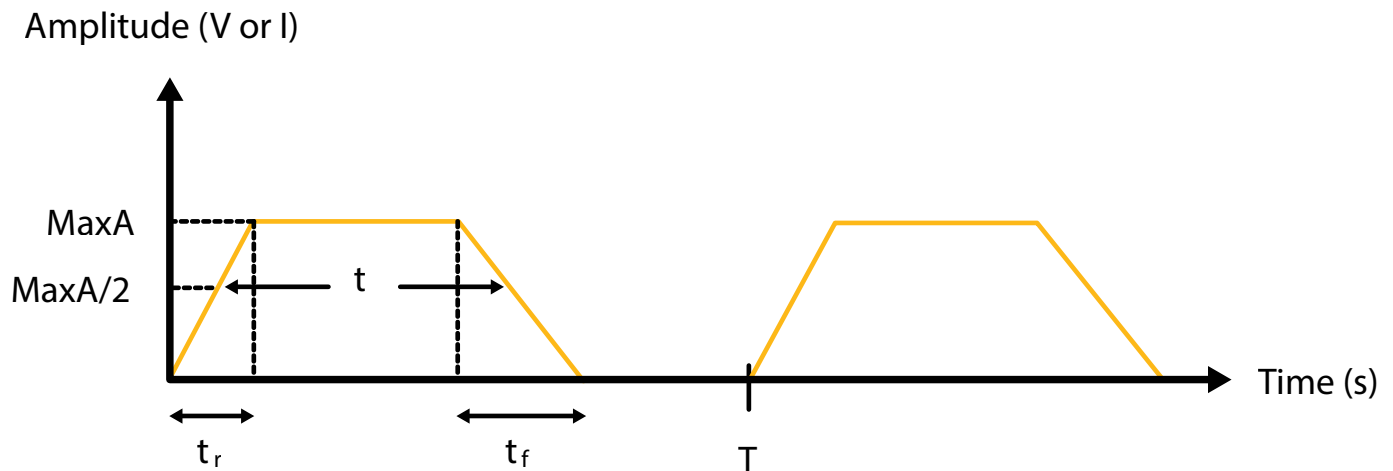
We know fast switching creates high-frequency noise, and we see those sharp peaks in the emissions plot. Now, we will look closer at why a square or trapezoid signal actually produces those peaks.

We check the signal two ways: how it changes over time (time domain), and what high-frequency parts show up (frequency domain).

To control EMI, it is essential to understand how switching signals behave in the time domain and how those waveforms translate into frequency-domain noise.

Time-domain parameters of switching waveforms

Any repeated square wave or trapezoidal signal can be described as below:



Parameters that define a square/trapezoid wave (time-domain).
Image credit: EMC United.

Where:

- T = Period of the signal (sec): time required for one complete cycle
- τ = Duty cycle (sec): time the signal is HIGH
- τ_r = Rise time (sec): time to go from LOW to HIGH
- τ_f = Fall time (sec): time to go from HIGH to LOW

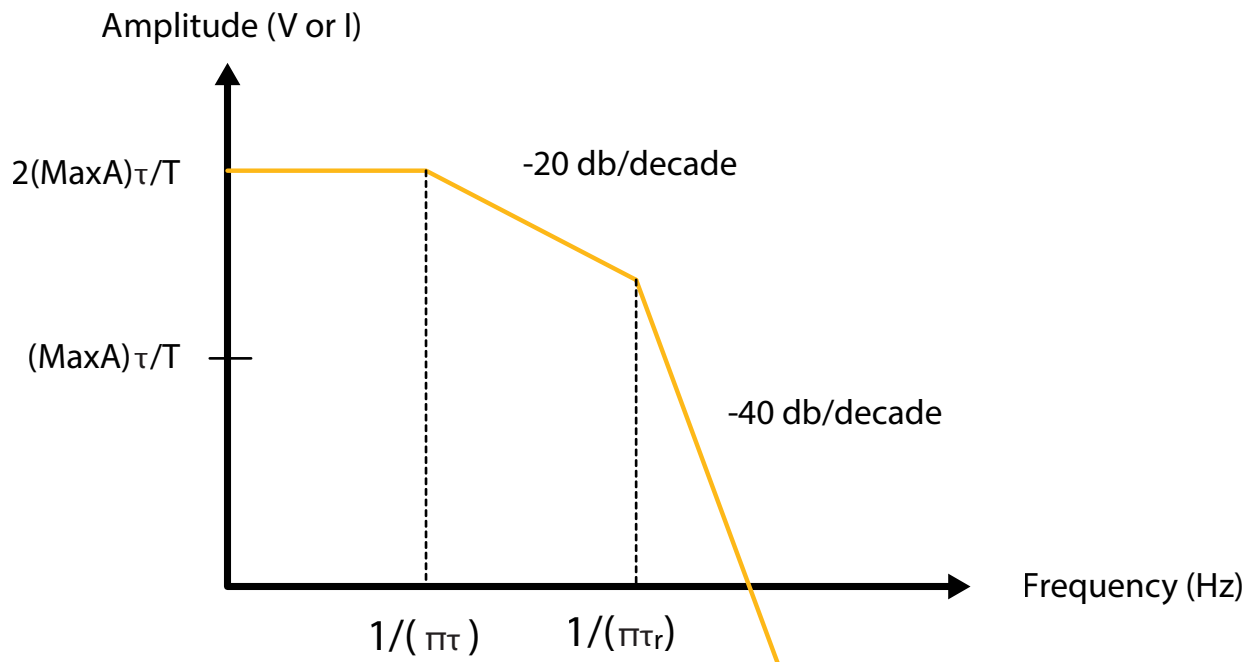
These parameters describe the exact shape of the switching signal as you see it on an oscilloscope.

- Faster rise or fall time means sharper edges → more high-frequency noise gets created.
- Slower rise or fall time means gentler edges → less high-frequency noise.

For instance, a shorter period (T) means a higher fundamental frequency, and asymmetric rise and fall times can introduce even more harmonics. Understanding them helps predict EMI issues early in the design phase.

Frequency-domain parameters of switching waveforms

These parameters determine the maximum amplitude envelope of harmonic noise in the frequency domain.



Fall off of harmonic noise terms in the frequency domain.
Image credit: EMC United

Key points about the graph:

1. Harmonic noise begins to decrease after the frequency $(1/\pi*\tau)$.
 - This is related to the duty cycle.
2. Harmonic noise decreases much faster after the frequency $(1/\pi*\tau_r)$.
 - This is related to the rise time.
3. If you increase τ_r (i.e., make the rise time slower), the second breakpoint moves to the left.
 - A lower frequency breakpoint leads to less high-frequency noise.
 - This generally improves EMC performance.
4. Signals with greater amplitude will also generate louder noise.
 - When looking for potential noise sources, look for circuits using the most energy/power/current, regardless of voltage.
 - Clocks, which service multiple parts of a board, will use more power than an individual IC pin and are more likely to cause EMC problems.
 - Power electronics that use switching operations are often the worst offenders for this reason.

For instance, a 20 MHz clock with 2 ns rise time will have significant harmonic content up to at least $1/(\pi \times 2 \text{ ns}) = 159 \text{ MHz}$. The same clock with a 10 ns rise time may only have significant harmonics up to about 32 MHz, a major improvement for EMC.

This example shows that just making the rise time a little slower cuts the high-frequency noise way down.

Effect of asymmetric rise and fall times

The time domain graph above assumes rise time and fall time are equal. If they are different, the worst case (greatest amplitude of high-frequency noise) is determined by whichever time is shorter, rise or fall.

High-order harmonics can still cause problems

Even when high-order harmonics have very small amplitudes, they can still cause EMC failures.

Why does this happen?

A weak, high-frequency noise signal can find a structure that acts as an antenna at that exact frequency. This combination can create a significant EMC problem.

For instance, a 100 MHz clock has 25th harmonic at 2.5 GHz. Even though the 25th harmonic is very weak, if there is a cable or trace that happens to be exactly the right length to radiate 2.5 GHz efficiently, you could have a radiated emissions problem. More information on radiating structures (antennas) is covered in section 4.

3.2 Pulse-width modulation (PWM)

PWM is a technique used to control variable outputs by rapidly switching power on and off. Common applications include:

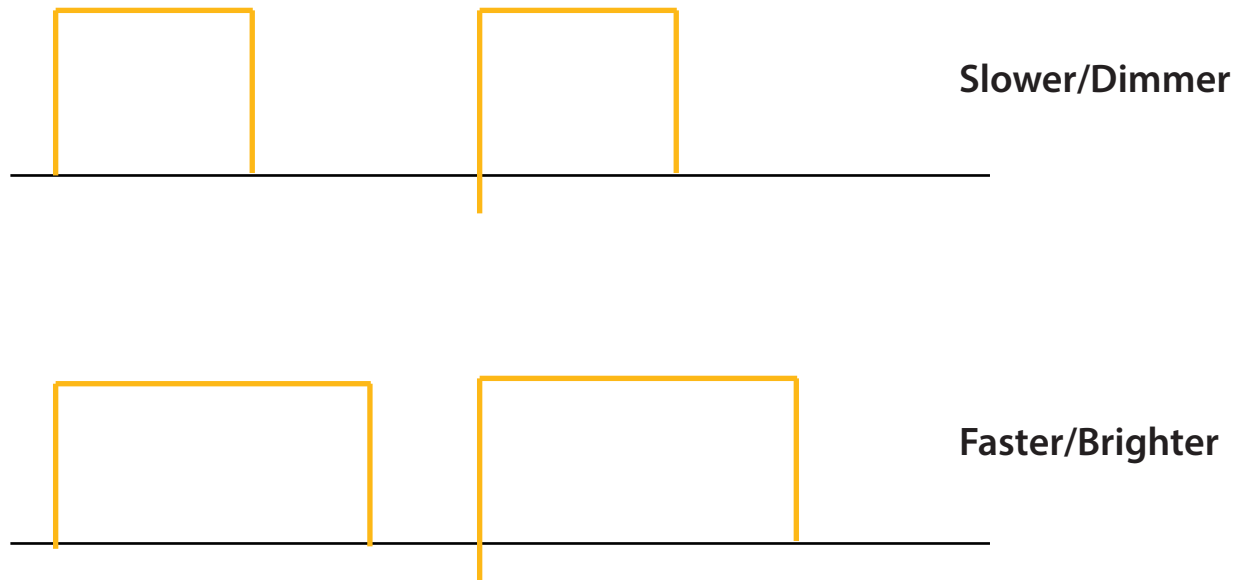
- LED brightness control
- Motor speed control
- Heating element temperature control

3.2.1 How PWM works

Power is switched on and off repeatedly. The percentage of time the power is ON (duty cycle) determines the output level.

Examples:

- **30%** duty cycle = dimmer light or slower motor
- **80%** duty cycle = brighter light or faster motor



Pulse-width modulated control signals. Image credit: EMC United

3.2.2 EMI from PWM signals

As with other switching operations discussed earlier, PWM creates high-frequency noise through rapid on/off transitions. The amount of EMI depends on:

- Rise and fall times
- PWM frequency
- Length of wires or traces carrying the PWM signal
- Amount of energy/current being switched

3.3 Motors

Motors are another common source of EMI. Motor types that create different noise patterns include:

- Brushed DC motors
- Brushless DC motors
- AC motors

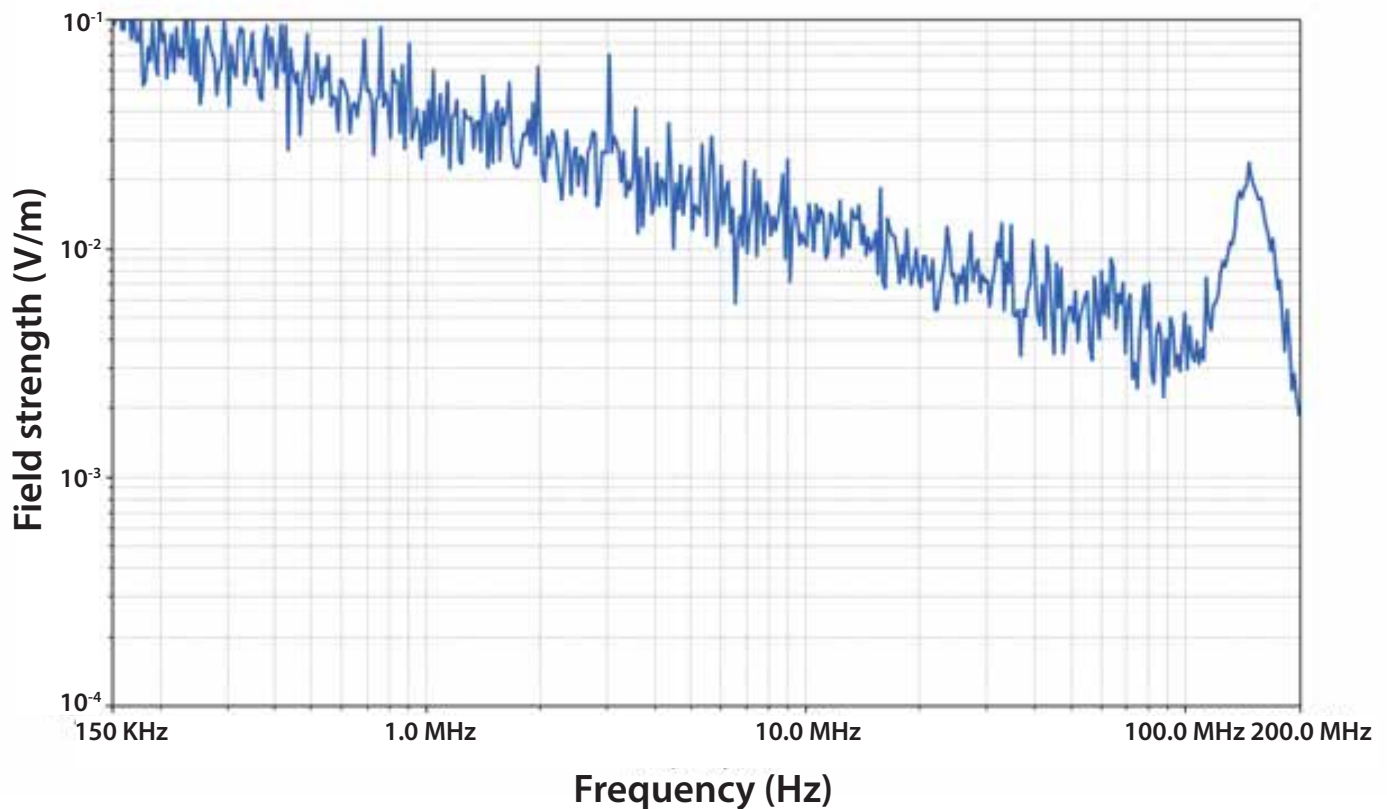
Noise frequency range

Motor noise typically occurs in the range of 100 kHz to 200 MHz.

Noise type

Motor noise is usually broadband noise. This means:

- The noise does not have sharp, distinct frequency peaks.
- Energy is spread across a wide frequency range.
- The noise spectrum looks more like a "hump" than individual spikes.



Illustrative radiated electric field from an unfiltered brushed DC motor. Image credit: EMC United

Why motors create broadband noise

Motors create broadband noise because of:

- Arcing at brushes (in brushed motors)
- Commutation events (switching in motor windings)
- Arcing at bearings
- Magnetic field variations

4. What structures cause radiated EMI in PCBs?

Radiated interference occurs when high-frequency noise couples into physical structures (traces, cables, slots, or enclosures) that act as unintended antennas. These structures are never intended to radiate and become the cause of EMI.

The most common accidental antennas are discussed below:

4.1 Cables

Cables can make very effective monopole antennas, sometimes even when they are shielded. If cable shields are not implemented correctly, they can become antenna elements themselves (see section 5.6).

When do cables radiate?

Cables are the best antennas when their length is close to $\lambda/4$ (one quarter wavelength) of a particular frequency. At this length, the cable resonates, converting conducted noise into strong radiated fields.

For instance, let's say you have a 20 MHz clock signal:

- The 5th harmonic of 20 MHz is 100 MHz
- The wavelength of 100 MHz is 3 meters
- One quarter wavelength = $3 \text{ m} \div 4 = 75 \text{ cm}$

A cable that is 75 cm long will be an excellent antenna for the 5th harmonic of your 20 MHz clock. This is true even if the 5th harmonic is much weaker than the fundamental 20 MHz signal.

Internal cables matter too

Unintended antennas are not limited to external cables. Internal wiring and board-to-board interconnects can radiate as well, especially in the following forms:

- Cables connecting daughterboards to motherboards
- Mezzanine board connections
- Flexible flat cables/ribbon cables
- Wire harnesses inside enclosures

Both daughterboard and mezzanine configurations can easily become antenna structures when:

- The connection length is problematic for the frequencies involved.
- Ground connections between boards are poor.

Design recommendations for board-to-board connections

- Ensure ground connections are well-established between different boards.
- Make connections as short as possible.
- Use multiple ground connections, not just one or two.
- Consider the frequencies in your design and avoid cable/connection lengths near $\lambda/4$.

4.2 Ribbon cables

Ribbon cables frequently cause EMI problems in electronic systems. They are naturally transmission lines because they have multiple parallel conductors carrying high-frequency signals.

However, their configuration determines whether they control electromagnetic fields effectively or create EMI problems. The most common interference issues arise from how return paths are implemented within the ribbon cable. (See section 5.1 for more on cables.)

Let's see what causes interference issues in ribbon cables and how to fix them.

4.2.1 Poor ribbon cable configurations

Many ribbon cables use only a single ground or return wire located at one edge of the cable.



A ribbon cable with a single ground. Image credit: EMC United

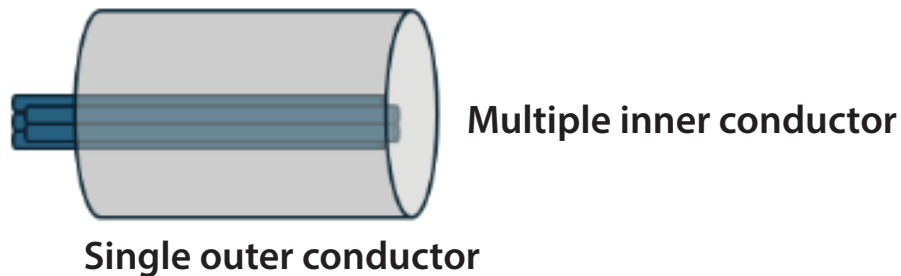
Why does this cause problems?

When multiple signal wires share a single ground/return wire:

- The electromagnetic fields from each signal are forced to overlap.
- Fields interact with each other (crosstalk).
- Fields are not well contained.
- The cable can radiate.

Using a ribbon cable with one ground wire is like trying to build a coaxial cable with:

- Multiple center conductors (each carrying a different signal).
- Only one outer conductor trying to contain all the fields.



Coaxial cable configuration with a single ground. Image credit: EMC United.

This does not work well. The fields from multiple signals will tend to interfere with each other. To reduce radiation and crosstalk, the return paths must be distributed so that each signal's EM fields are properly contained.

4.2.2 Better ribbon cable configurations

Best practice: Dedicated ground for each signal

The most effective configuration is to provide a dedicated ground/return wire for each signal wire.



Ribbon cable with alternative signal and ground wires offers better field containment. Image credit: EMC United.

Implementation:

- Alternate signal and ground wires throughout the ribbon cable.
- Pattern: Ground-Signal-Ground-Signal-Ground-Signal...
- Make sure every signal has return current paths on both sides.

Why this works:

Each signal creates a transmission line with its adjacent ground wires:

- Fields are contained between signal and adjacent ground wires.
- Minimal overlap between different signal fields.
- Much less crosstalk.
- Significantly reduced radiation.

Alternative: Multiple symmetrically-placed grounds

If you cannot dedicate a ground wire for every signal, use multiple ground wires distributed throughout the cable.



Multiple ground wires are distributed throughout the cable. Image credit: EMC United.

Implementation:

- Place ground wires symmetrically throughout the ribbon
- Example for 10-conductor cable:
Ground-Signal-Signal-Ground-Signal-Signal-Ground-Signal-Signal-Ground
- Avoid placing all grounds on one edge.

Why this helps:

While not as effective as alternating signal-ground, this approach:

- Provides multiple return paths distributed across the cable.
- Reduces field spreading compared to a single ground.
- Improves EMC performance significantly over single-ground designs.

The small cost increase for additional ground conductors is far less than the cost of failing EMC testing and redesigning your system.

Ribbon cable rule: The worst EMC performance comes from ribbon cables with all signals grouped together and ground wires grouped at one edge. Instead, distribute ground wires throughout the cable.

4.3 Traces

PCB traces can also be antennas. The same antenna principles that make cables radiate apply to signal tracks, especially when their physical length approaches a fraction of the signal wavelength.

To understand when traces become problematic, let's look at a practical example:

Let's say you have a 500 MHz clock signal:

1. The 5th harmonic of 500 MHz is 2.5 GHz
2. The wavelength of 2.5 GHz is 12 cm
3. One quarter wavelength = $12 \text{ cm} \div 4 = 3 \text{ cm}$

An unterminated 3 cm trace carrying the 2.5 GHz harmonic could be a radiating antenna.

Design recommendations for traces

1. Keep high-frequency traces as short as possible.
2. Properly terminate transmission lines.
3. Route traces over solid reference planes (discussed in section 5.2).
4. Avoid routing high-frequency signals near board edges.

4.4 Ungrounded components

Not all radiating structures are signal-carrying conductors; large ungrounded metal parts can also behave as unintended antennas. Heat sinks are a common example.

Why heat sinks radiate

When a heat sink is not connected to ground (or a reference plane), it can:

- Act as a metal antenna element.
- React to high-frequency signals from nearby components.
- Radiate those signals into the surrounding space.

Design recommendation

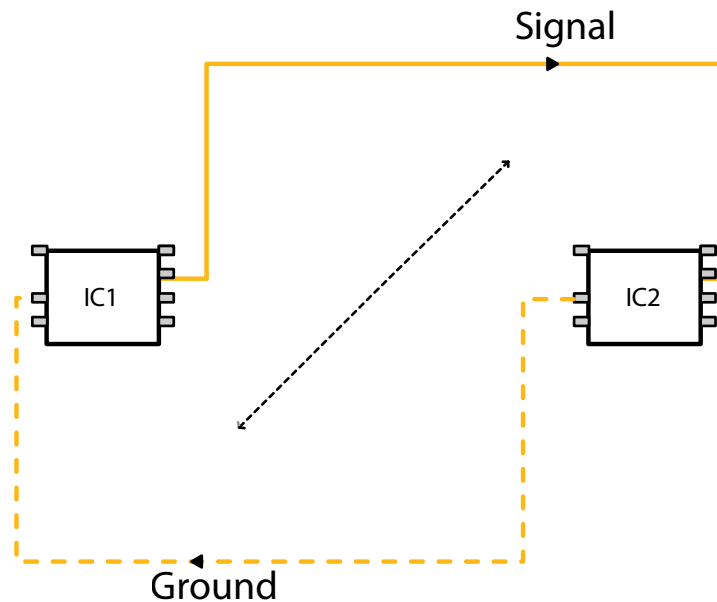
Connect large metal components (especially heat sinks) to a ground or reference plane. Use:

- Screws to mounting pads
- Thermal interface material with conductive properties
- Grounding clips or springs

4.5 Loops

In addition to cables, traces, and metal structures, the geometry of current flow itself can form efficient radiating structures.

Every time current travels from a source to a load and back to the source, it forms a current loop, and it creates a magnetic field. That magnetic field, and its associated electric field, can radiate. Every current loop is a potential antenna.



Example loop of a signal and ground trace on a PCB. Image credit: EMC United

Common sources of current loops:

1. Signal trace with distant return path.
2. Power and ground traces that enclose a large area.
3. Split reference planes, forcing the return current to take a long path.

Why loops are problematic

The larger the loop area, the more effectively it radiates. Loop radiation increases with:

- Frequency (higher frequency = more radiation)
- Current amplitude (more current = more radiation)
- Loop area (bigger loop = more radiation)

Design recommendations

1. Minimize loop areas wherever possible.
2. Route signal and return traces close together.
3. Provide continuous reference planes for the return current.
4. Don't split ground planes unless absolutely necessary.

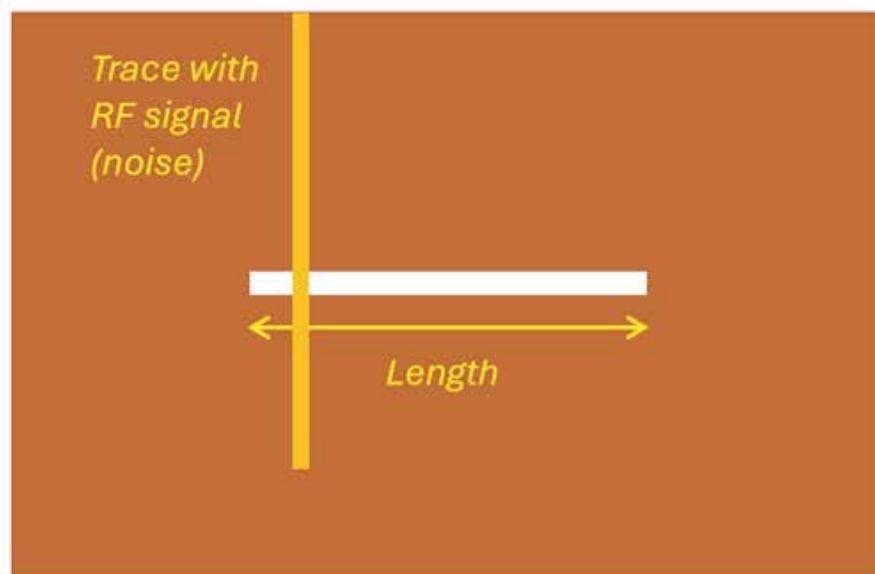
App notes will often be misleading in this regard, just because one recommends split planes does not mean the design will be successful in EMC testing.

4.6 Slots and apertures

Gaps in conductive surfaces, whether on PCBs or enclosures, can act as antennas. Understanding how slots radiate is essential for controlling emissions.

4.6.2 Slot antennas on PCBs

A slot antenna can be intentionally designed by running a microstrip line over a gap in a plane.



Notional slot antenna. Image credit: EMC United

Accidental slot antennas

In practice, most slot antennas on PCBs are not intentional. You can accidentally create a slot antenna when you route a trace:

- Over a gap or slot in an adjacent plane.
- Near an aperture in a conductive enclosure.

When slots radiate most effectively

Slot antennas are most effective when their longest dimension is $\lambda/2$ (half a wavelength).

For instance:

Let's say you have a 500 MHz clock signal:

1. The 5th harmonic of 500 MHz is 2.5 GHz
2. The wavelength of 2.5 GHz is 12 cm
3. Half wavelength = $12 \text{ cm} \div 2 = 6 \text{ cm}$

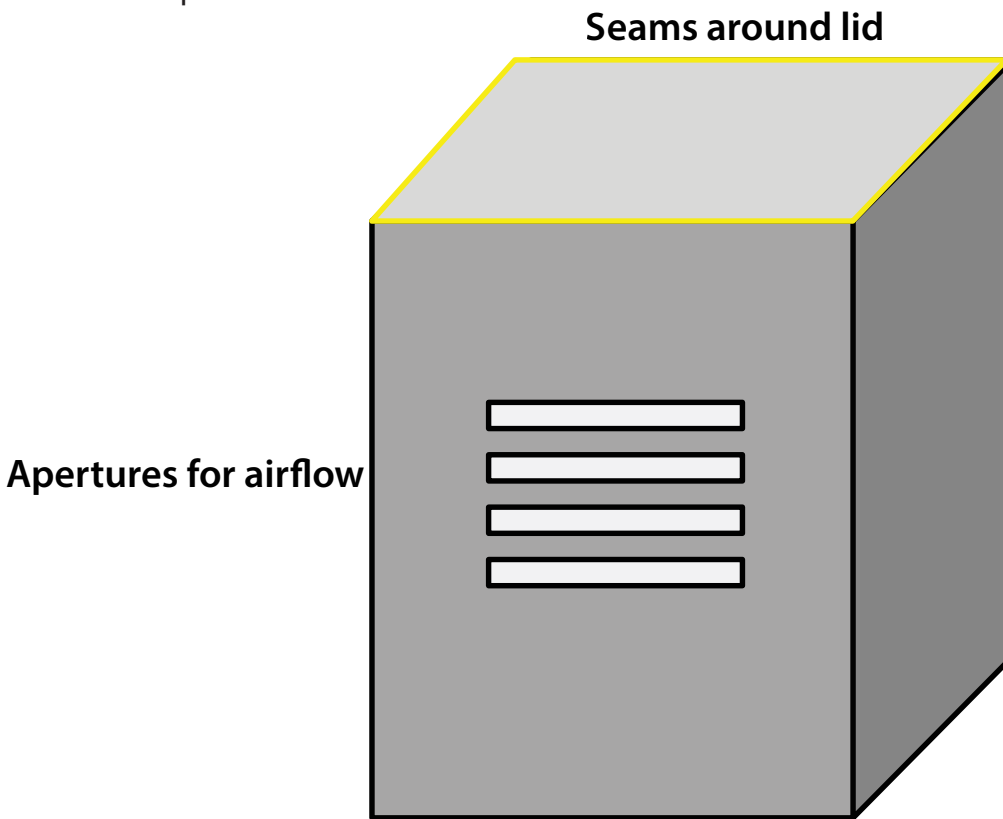
A 6 cm slot in a PCB ground plane could be an effective radiator if:

- There is a trace carrying 2.5 GHz noise nearby.
- The trace is routed over or near the slot.

4.6.3 Apertures in conductive enclosures

The same slot-antenna behavior applies beyond the PCB itself, especially at openings in metal enclosures. Metal enclosures need openings for various reasons:

- Ventilation holes for cooling
- Display windows
- Connector openings
- Seams between panels
- Access panels



Apertures in a conductive enclosure. Image credit: EMC United

How enclosure apertures cause problems

Any aperture can radiate as a slot antenna when:

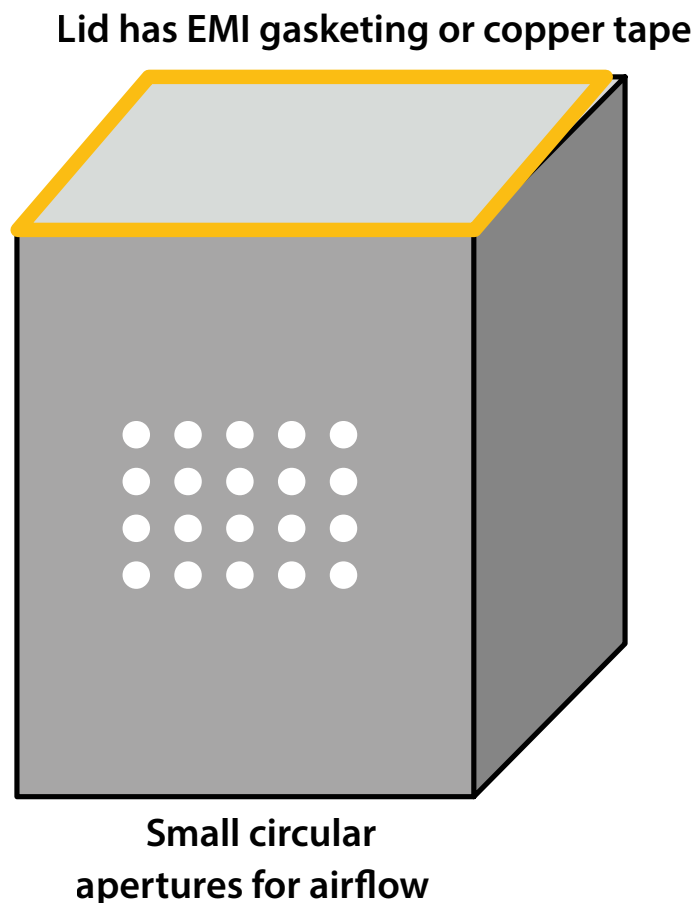
- EM energy exists inside the enclosure at the aperture's resonant frequency.
- External fields at that frequency interact with the aperture.

The combination of PCB traces or cables inside the enclosure and apertures in the enclosure can create radiated emissions problems.

Design strategies for apertures

Since most enclosures require some openings, the goal is to minimize their radiation effectiveness.

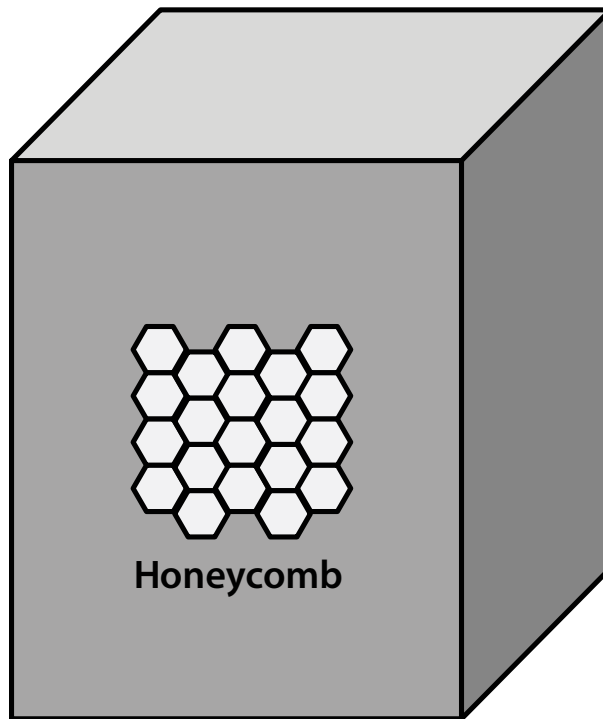
1. Replace long slots with small holes. Long, thin slots radiate more than small circular holes.



Enclosure with small circular apertures. Image credit: EMC United

2. Design options (from best to worst EMC performance):

1. Honeycomb mesh: The best option



Enclosure with honey-comb apertures. Image credit: EMC United

- Hexagonal pattern limits maximum aperture dimension
- Metal thickness adds additional filtering/attenuation
- Maintains good airflow
- Most expensive option

2. Many small circular holes: Good option

- Much better than slots
- Easier to manufacture than honeycomb
- Still provides adequate ventilation

3. Long thin slots: Worst option

- Most likely to create radiated emissions problems
- Avoid when possible

Seal removable panels

For panels removed rarely:

- Use copper tape around the perimeter to seal the gap between the panel and the enclosure body. Make sure the copper tape is conductive on both sides, including the adhesive.

For panels removed routinely:

- Use EMI gasketing material that is conductive and made of flexible material. It fills the gap between the lid and the enclosure. Examples: copper beryllium finger stock, conductive elastomer, and conductive foam.

Important: The gasket must make good electrical contact with both the lid and the enclosure body.

4.7 Electrical bonding

Bonding means creating a low-impedance electrical connection between two conductive surfaces. Good bonding is essential to ensure EMC.

4.7.1 Critical requirement: Direct metal-to-metal contact

Any conductive EMC mitigation methods (copper tape, gaskets, bond straps) must make direct contact with bare metal. They will not work effectively if:

- The metal surface is painted.
- The metal has a non-conductive surface treatment (like anodization).
- Is there any non-conductive coating between the mitigation and the metal?

Common bonding mistake

Mechanical connection $\neq$ Electrical connection

Fasteners (screws, bolts, and rivets) join two pieces together mechanically. However, this does not guarantee a good electrical connection.

For example, if two painted metal panels are joined using screws, they hold the panels together mechanically, but the paint prevents good electrical contact. The bond is poor and unreliable.

Problems with fastener-only bonding

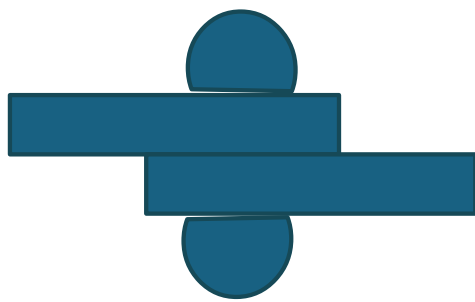
1. **Unreliable contact:** Paint, anodizing, or corrosion blocks electrical contact.
2. **Fasteners loosen over time:** Vibration, thermal cycling, and normal use cause fasteners to loosen gradually.
3. **Increased impedance as fasteners loosen:** As screws loosen, the electrical connection degrades.
4. **EMC failures develop over product lifetime:** A unit that initially passes EMC testing may fail later as fasteners loosen.

4.7.2 Proper bonding methods

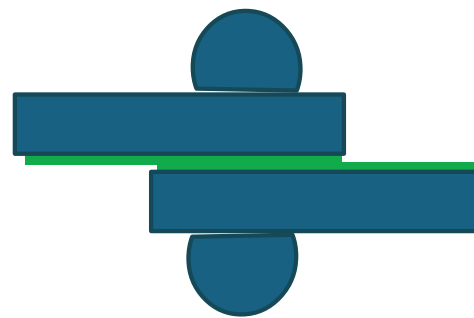
1. Best practice: Overlapped bare metal surfaces

This is the most reliable bonding method:

1. Remove paint or coating from the contact area.
2. Place bare metal surfaces directly against each other.
3. Hold them tightly with fasteners.
4. Ensure no non-conductive material is between the surfaces.



Metal-to-metal contact,
maximum effectiveness



Paint impeding metal-to-metal contact,
less effectiveness

Bonding works best with bare metal-to-metal contact. Image credit: EMC United

Why this works:

- Large contact area between bare metal surfaces.
- Low impedance connection at all frequencies.
- Contact is maintained even if an individual fastener loosens slightly.

2. Acceptable practice: Bond straps

Sometimes you cannot directly bond two metal surfaces. In these cases, use a bond strap (a flat metal strap or braid).

Requirements for effective bond straps:

1. Both ends must have bare metal contact

- Remove paint or coating where each end connects.
- Ensure full metal-to-metal contact at both ends.

2. Bond straps should be short and wide

- Ideal ratio: length no more than 5 times the width.
- Example: A 5 cm long strap should be at least 1 cm wide.
- This is not always possible, but aim for this ratio when feasible.

3. Avoid thin wires as bond straps

- Very thin wires add significant inductance.
- Inductance increases impedance at high frequencies.
- The goal is to have the lowest possible impedance at all frequencies, especially high frequencies where EMC problems occur.

Note: Some surface treatments, such as anodization, destroy surface conductivity. So two aluminum pieces may appear to be well bonded together, but if one or both of them is anodized, the bond will not be effective for EMC purposes.

Good bonding requires bare, conductive metal contact. If your enclosure is painted or anodized for appearance, plan bonding areas during the design phase. You may need to mask certain areas during finishing, or remove coating from specific bonding locations after finishing.

5. Design guidelines to ensure EMC in your PCBs

To achieve electromagnetic compatibility, your design must contain EM fields, provide uninterrupted return paths, suppress noise at its source, and prevent interference from escaping through conductive features.

Stick to these rules for a low-noise design:

5.1 Use transmission lines for field containment

Your PCB will have significant amounts of high-frequency noise on many traces. How can we prevent this noise from becoming an EMI or signal integrity problem?

The solution: Provide transmission lines

Incorporate transmission lines to route high-frequency signals. This contains the EM fields so they cannot escape and cause problems.

Transmission lines are simpler than you think

Creating transmission lines for EMC does not need to be complicated. Most common types include:

1. **Microstrip:** A trace over a solid plane.
2. **Two-wire transmission line:** Two traces routed next to each other (one carries signal or power, the other carries return current).
3. **Stripline:** A trace buried between two solid planes.

Important note for EMC

For EMC purposes, these transmission lines do not need strict controlled impedance requirements. It is important for:

1. Signal integrity
2. RF circuits
3. High-speed digital signals

But for EMC field containment, we need the transmission line structure to exist. The structure contains the fields and prevents them from escaping.

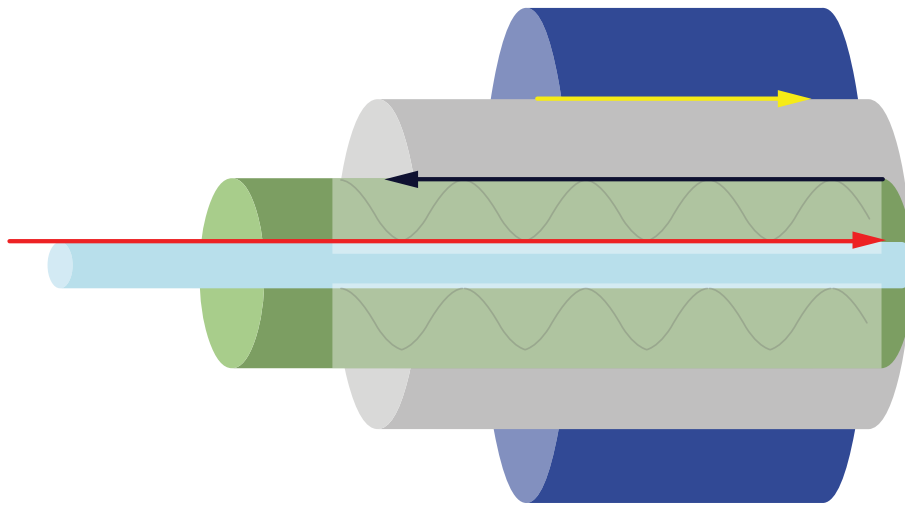
5.1.1 Understanding field containment: The coaxial cable example

Coax confines fields perfectly.

For instance, a cable can simultaneously carry:

- DC power (phantom power)
- High-frequency RF signals (like AM/FM radio)

Both signals travel along the same cable without interfering with each other or radiating into the environment.



Cross-sectional view of a coaxial cable. Image credit: EMC United

When a coax is operating ideally:

- Current flows out on the outer surface of the inner conductor (Red).
- Current returns on the inner surface of the outer conductor (Black).
- Energy is carried by the electromagnetic fields in the dielectric.
- Noise remains on the outside of the outer conductor (Yellow).

A properly implemented coaxial cable provides nearly perfect containment of EM fields.

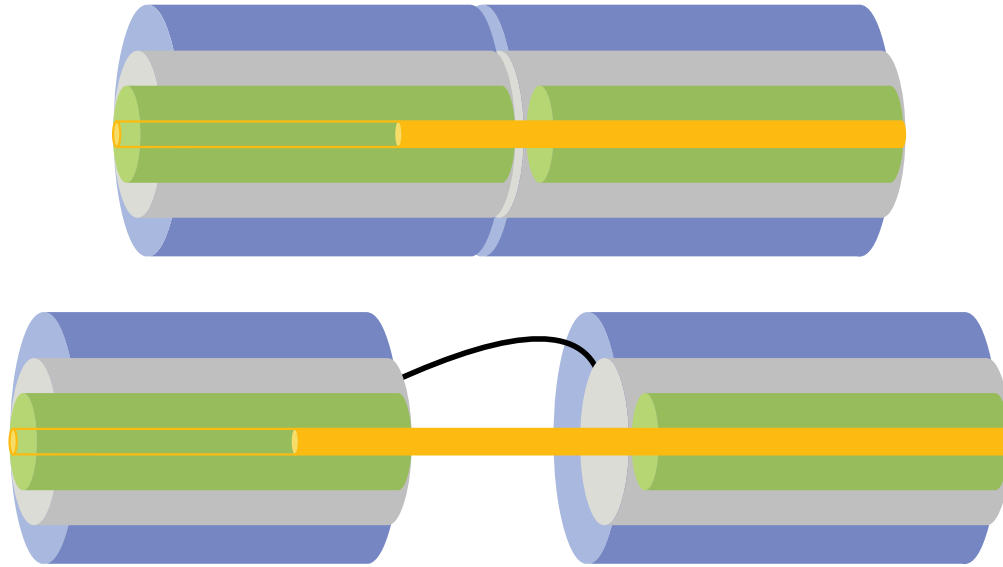
As frequency increases:

- The signal experiences more attenuation (it gets weaker as it travels along the cable length).
- But the fields remain contained inside the cable.

Field containment works regardless of frequency, from DC through GHz.

How not to connect coaxial cables

We intuitively understand that one should never connect a coaxial cable with a pigtail or drain wire.



Different methods of connecting two coaxial cable sections. The bottom one should never be used. Image credit: EMC United.

Why the pigtail method is wrong

The thin wire connection (i.e., the pigtail method) technically allows current to flow continuously on the outer conductor. However:

- The fields are not contained.
- Energy leaks from the cable at the connection point.
- The cable becomes an ineffective shield.
- EMI and signal integrity problems result.

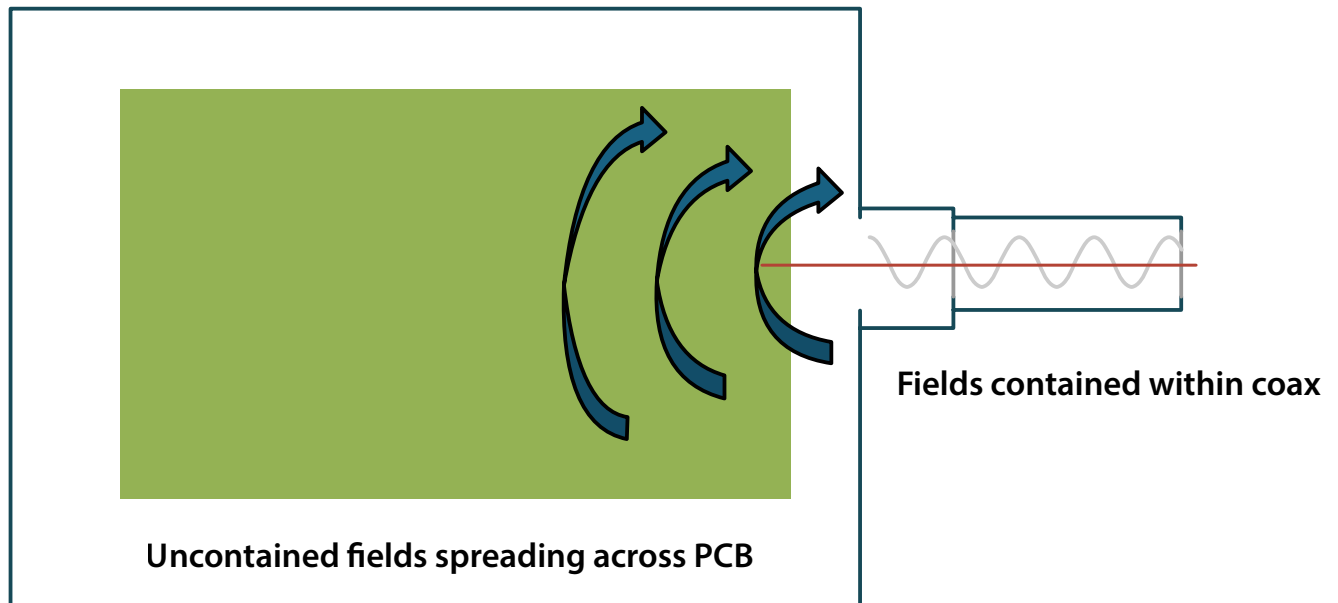
The lesson: Proper field containment requires proper connection methods, not just a conductive path.

Bringing coaxial cables into enclosures

The same principle of maintaining containment applies when bringing a coaxial cable into a metal enclosure or connecting it to a PCB.

Poor implementation: No field control

Bringing a coaxial cable into a metal enclosure without considering field containment:



Coax feeding a PCB in a metal enclosure with no field control. Image credit: EMC United

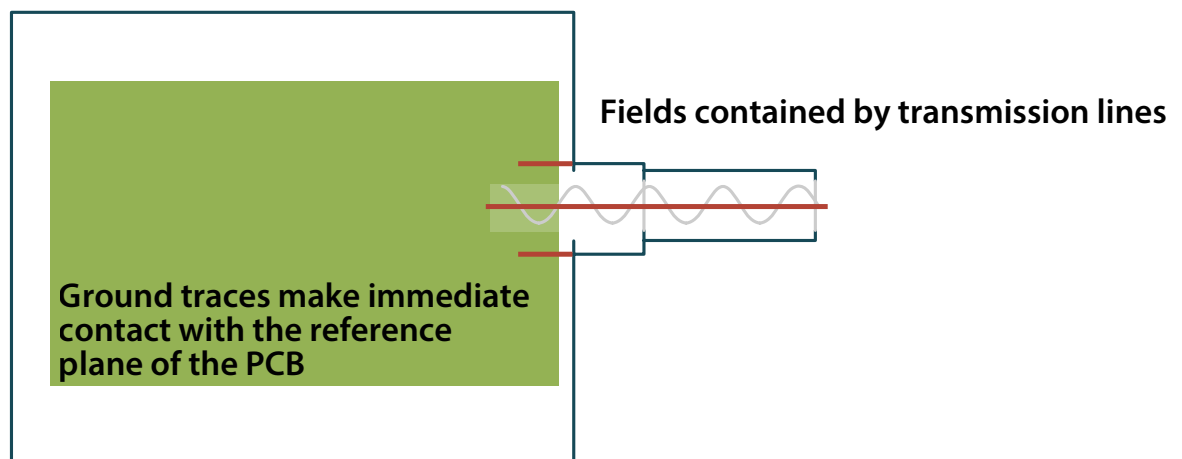
What's wrong?

If no provision is made for field containment:

- Fields escape from the cable.
- They spread throughout the interior of the unit.
- They can interfere with other circuits.
- They can radiate from apertures in the enclosure.

Better implementation: Continue the transmission line

A better approach is to continue the transmission line structure into the unit:



Improved coax connection with PCB. Image credit: EMC United

How this works?

1. The coaxial cable provides 360° field containment (cylindrical).
2. At the transition, the center conductor continues as a PCB trace.
3. The outer conductor connects to two symmetric ground traces.
4. These ground traces immediately connect to the circuit board reference plane.

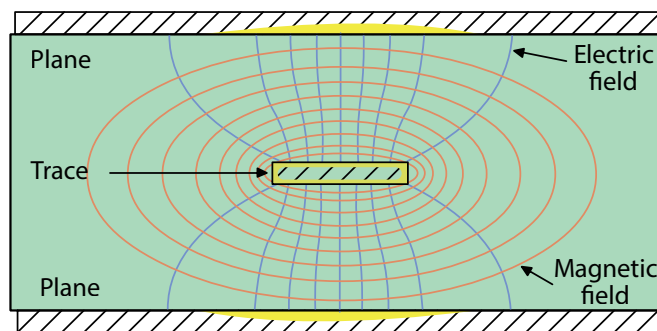
Result: Field containment is maintained as the coaxial line transitions to the interface with the board.

Circuit board transmission lines don't need to be perfect. Very few board-level transmission lines can achieve perfect field containment. But ideal confinement is rarely necessary. Getting close is usually enough for good EMC performance.

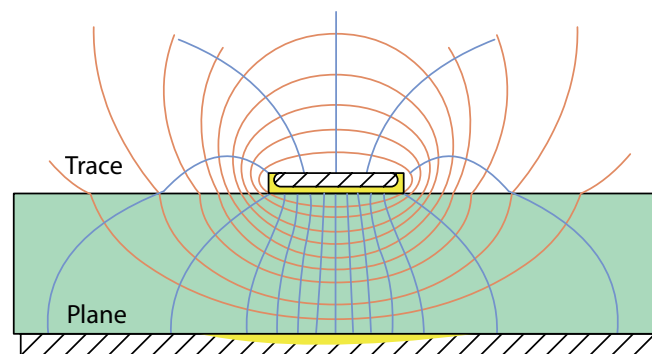
5.1.2 Choosing between stripline and microstrip

We understand that coaxial cable is ideal for field containment. Let's now see how stripline and microstrip help you contain fields.

Field distribution in different transmission line types:



Stripline transmission line



Microstrip transmission line

Different transmission lines using PCB traces. Image credit: Rick Hartley

Stripline (buried trace)

- Trace is sandwiched between two solid planes
- Offers better field containment
- Fields are more tightly confined
- Best choice for particularly sensitive or especially noisy traces

When to use stripline:

- High-frequency clock signals
- Sensitive analog signals
- Very noisy switching signals

Microstrip (surface trace)

- Trace on the outer layer with a plane on the adjacent layer
- Some fields extend above the plane and beyond the plane edges
- More than adequate for most applications
- Easier to route and debug

When to use microstrip:

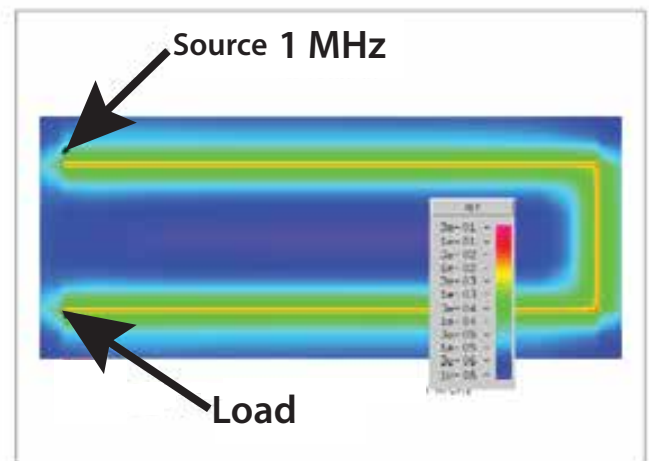
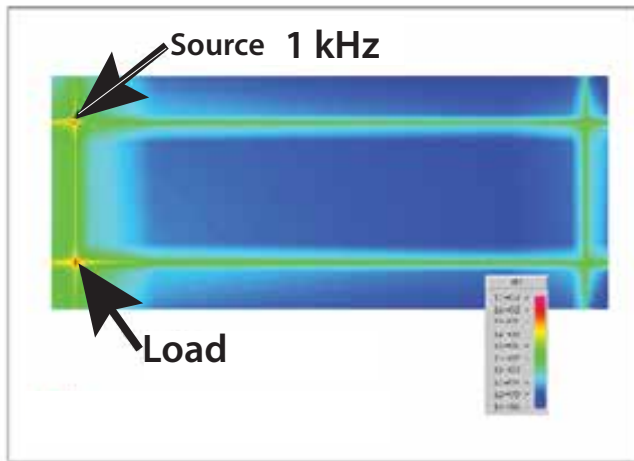
- Most digital signals
- Power delivery
- General-purpose I/O
- Medium-frequency signals

Natural field confinement at high frequencies

At higher frequencies, field behavior changes in a helpful way for EMC.

Higher frequency = better self-containment

As frequency increases, electromagnetic fields naturally confine themselves closer to the trace and plane.



Return current density underneath a continuous trace.
Image credit: Bruce Archambeault

What this means:

- High-frequency fields stay close to the trace.
- They don't spread over large areas of the plane.
- Return current flows in a narrow path directly under the signal trace.

Important consequence:

This natural field confinement minimizes the need to split reference planes. In modern electronics:

- You generally do not need to split analog and digital grounds.
- The fields contain themselves.
- Splits often cause more problems than they solve.

DON'T split ground planes to separate analog and digital sections.

DO use continuous ground planes; let natural field confinement do the work.

Transmission lines on single-layer boards

On a single-layer board (no ground plane), you have limited options:

Option 1: Co-route signal and return

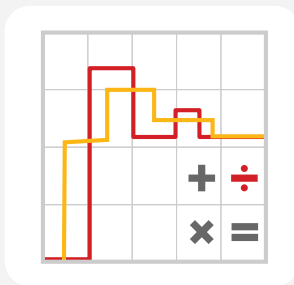
Route the signal trace and its return trace close together, parallel to each other. This creates a two-wire transmission line.

Option 2: Guard traces

Place guard traces on both sides of a signal trace (parallel). Connect these traces to ground at multiple points. This provides some field containment.

Single-layer boards have inherent EMC challenges. The techniques above help, but they cannot match the performance of multilayer boards with solid planes.

EMC design rule: Every high-frequency signal, including noise, needs a transmission line structure. This means every trace carrying high-frequency signals must have an adjacent reference plane, or must be routed with its return path close by.



PCB DESIGN TOOL

Transmission Line Reflection Calculator



TRY TOOL

5.2 Route traces to minimize radiation and coupling

PCB routing directly affects how easily EM fields escape from traces or couple into other structures.

The following guidelines help you contain fields, reduce antenna effects, and improve overall EMC performance.

5.2.1 Have traces away from the board edges

Traces routed too close to the edge of the board can radiate effectively. This happens because there is not enough ground plane beneath the trace to contain the fields.

Design rule

Keep traces and components at least one dielectric thickness away from the board edge.

Why this matters

When there is insufficient plane beneath a trace:

- Fields escape from the board edge.
- The trace becomes a more effective radiator.
- Fields are more likely to interact with apertures in a metal enclosure.
- EMI problems result.

5.2.2 Keep traces as short as possible

Shorter traces have multiple benefits, including less chance for fields to affect other circuits, lower impedance, smaller loop area, reduced antenna efficiency, and better signal integrity.

Design rule

Always minimize trace length, especially for:

- High-frequency signals
- Clock signals
- Switching power supply traces
- PWM signals

5.2.3 Never leave pins or traces floating

Unterminated or floating traces and pins are serious EMC problems.

Design rule

Every pin and every trace must be:

- Connected to a defined circuit
- Terminated properly
- Connected to ground through an appropriate resistor

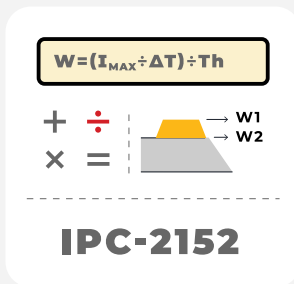
Why floating structures are dangerous

They can easily become antennas for high-frequency noise. They act as:

- Emitters: Picking up noise from nearby circuits and radiating it.
- Receivers: picking up external fields and conducting them into sensitive circuits.

Common mistakes to avoid

- Unused microcontroller pins left unconnected.
- Prototype traces left on production boards.
- Test points connected to nothing.



PCB DESIGN TOOL

Trace Width, Current Capacity and Temperature Rise Calculator



TRY TOOL

5.3 Design stack-ups with solid reference planes

Stack-up is critical for EMC success. It is one of the most important factors in passing EMC testing on the first attempt.

The fundamental requirement

Have sufficient solid planes to control all high-frequency fields produced by switching operations and digital signals.

Stack-up design tips to reduce EMI

5.3.1 Implement adjacent reference planes for every noisy layer

Any layer with a potentially noisy signal or power traces must have a solid plane immediately adjacent to it. The adjacent plane can be:

- A ground/return reference plane (most common)
- A power plane (also effective)

This adjacent plane provides immediate low-impedance return paths and contains fields.

On any multilayer board: Layer 2 should be a solid ground/return reference plane.

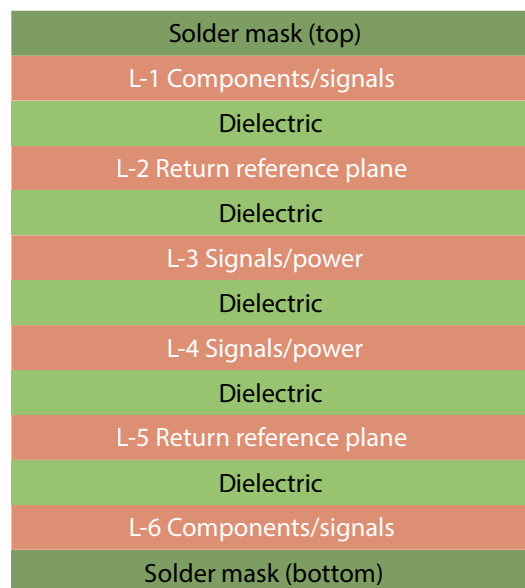
Why this is important

- Layer 1 (top) will have many components and traces.
- These components switch and create high-frequency noise.
- The solid plane on layer 2 provides an immediate return path.
- Fields are contained between layer 1 and layer 2.

Once adjacent planes are provided, the next factor that strongly influences EMC performance is the total number of layers.

Six-layer boards are excellent for EMC

Even multiples of three (6, 12, 18, etc. layers) make particularly robust designs for EMC.



Example of a good 6-layer board stackup. Image credit: EMC United

Typical 6-layer arrangement:

- Layer 1: Component/signal layer (adjacent plane on layer 2)
- Layer 2: Ground plane (solid)
- Layer 3: Signal/power (adjacent plane on layer 2)
- Layer 4: Signal/power (adjacent plane on layer 5)
- Layer 5: Ground plane (solid)
- Layer 6: Component/signal layer (adjacent plane on layer 5)

Additional field control: Ground floods on outer layers

For even better field control, add ground flood (also called copper pour) on the top and bottom layers.

Critical requirement

Ground floods only help if they have solid connections to an adjacent reference plane.

How to connect ground floods

- Multiple vias throughout the flood area.
- Connect directly to the adjacent ground plane.
- Use enough vias to ensure low impedance.

WARNING:

Floating metal (metal not connected to anything) can act as an antenna instead of a shield. A ground flood without good via connections may be worse than no ground flood at all.

Reference plane placement matters for 4-layer boards

With only four layers, the placement of planes is critical to maximize containment.



Alternate 4-layer stack-ups. The arrangement on the left is preferable. Image credit: EMC United

Better 4-layer stack-up (left)

- Layer 1: Components/signals (adjacent to ground on layer 2)
- Layer 2: Ground plane (solid)
- Layer 3: Power plane (solid)
- Layer 4: Components/signals (adjacent to power on layer 3)

Why this is better

- Both outer layers have adjacent planes.
- Good field control for most traces.

Poorer 4-layer stackup (right)

- Layer 1: Components/signals (adjacent to ground on layer 2)
- Layer 2: Ground plane (solid)
- Layer 3: Signal/power (adjacent plane on layer 2)
- Layer 4: Components/signals (with no adjacent plane, must be routed as if it were a single-layer board)

Why this is worse

- Layer 4 has no adjacent plane.
- Very little field containment for signals and noise on layer 4.

Routing on layers without adjacent planes

If a particular layer does not have a solid plane immediately adjacent, you must route that layer as if it were a single-layer board. Use the techniques discussed in the transmission lines section 5.1:

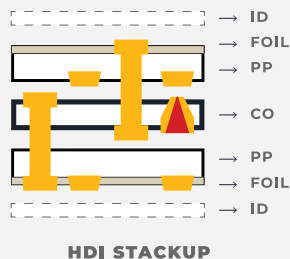
- Co-route signal/power and return traces.
- Use guard traces if necessary.

Why this matters

Without an adjacent plane, you have no automatic field containment. You must create transmission line structures manually through careful routing.

Cost vs. EMC performance

Adding layers to your PCB increases manufacturing cost. However, failing EMC testing and requiring board redesign may cost more. A 6-layer board that passes EMC on the first try is cheaper overall than a 4-layer board that requires multiple respins.



PCB DESIGN TOOL

Stackup Designer



TRY TOOL

5.4 Add stitching vias near layer transitions

When a signal transitions between layers using a via, the field containment can be disrupted. This allows fields to escape and potentially cause EMI.

The solution is to place ground stitching vias near signal vias that transition between layers.

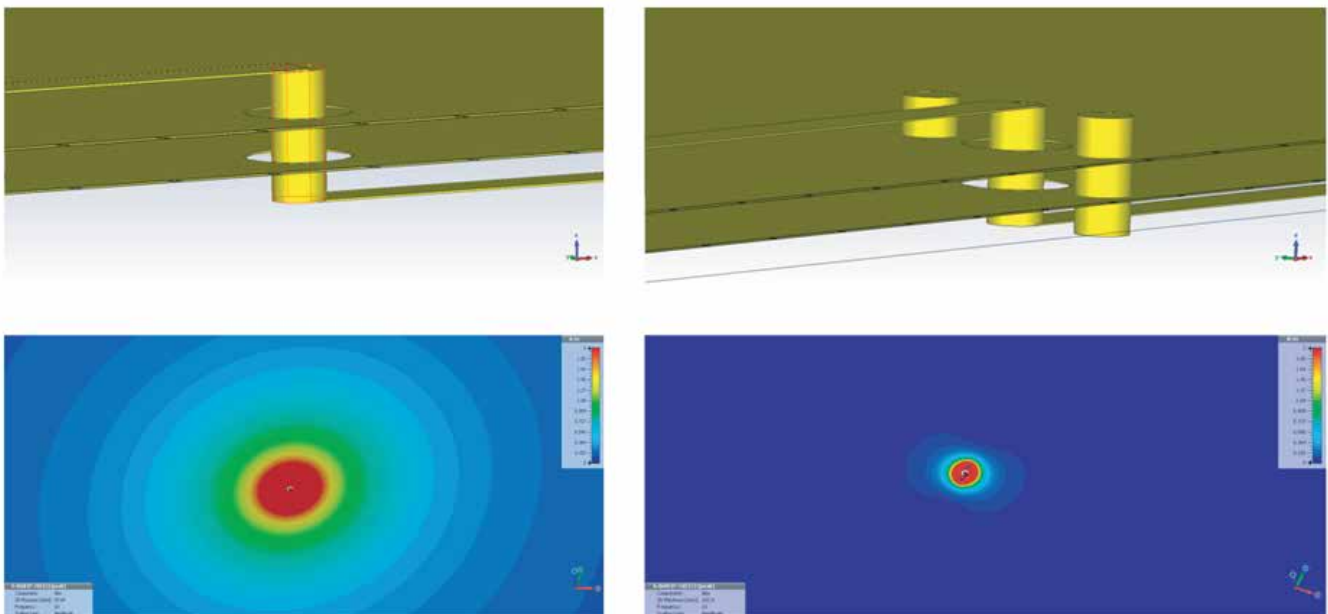
5.4.1 How stitching vias work

To understand why stitching vias are effective, let's look at how fields behave during a layer transition.

Creating a continuous transmission line

Stitching vias helps maintain field containment around a signal via. They provide a return path that follows the signal through its layer transition.

Effect of stitching vias



Effect of adding ground vias near a transitioning signal, simulating current density on the reference plane with and without stitching vias. Image credit: Scott Piper/CST

Without stitching vias:

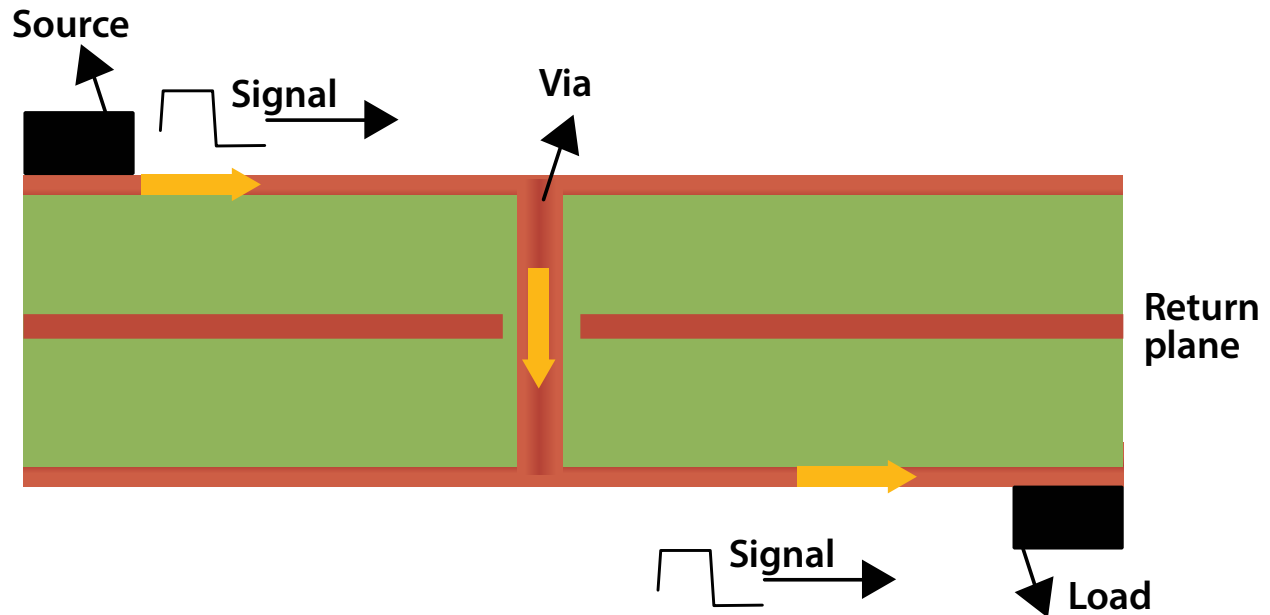
- Fields spread out around the via.
- Energy radiates into the surrounding board areas.
- Potential EMI problems might occur.

With stitching vias:

- Fields remain confined near the signal path.
- Return current has a nearby path to follow the signal.
- Much better EMC performance.

5.4.2 When stitching vias are not needed

If a signal only passes from one side of a plane to the other side of the same plane, stitching vias are unnecessary.



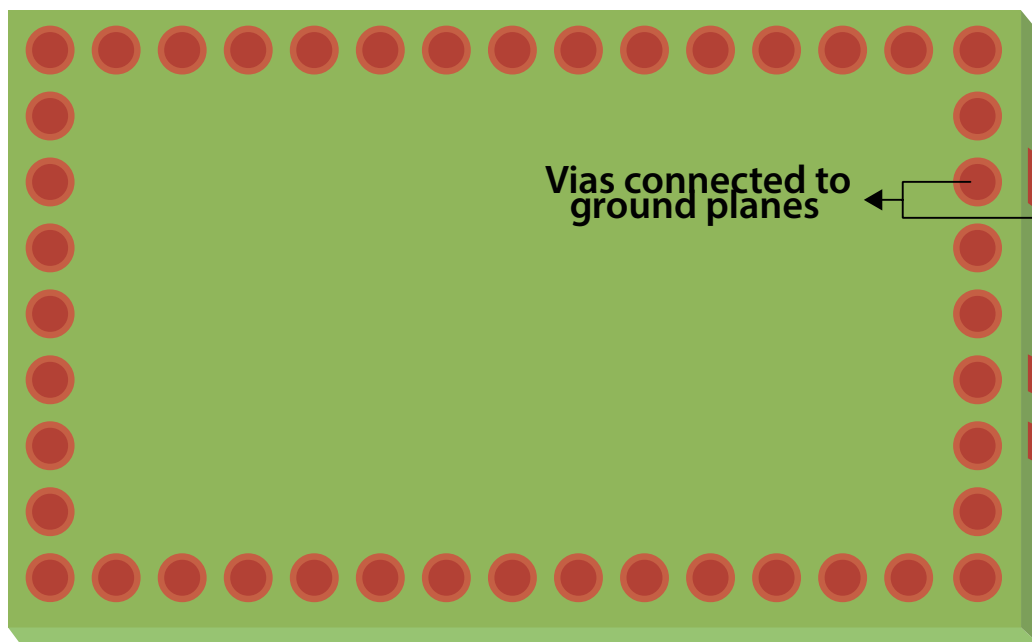
A via transitioning through a continuous ground plane does not require stitching vias, as the plane provides an uninterrupted return path for the signal.

In this case:

- The signal goes from layer 1 to layer 3.
- Layer 2 is a solid ground plane between them.
- The signal via passes through the plane.
- The solid plane provides a continuous return path.
- No stitching vias needed.

5.4.3 Via fences for field containment

A via fence is a row of grounded vias near the board edge. It can help prevent fields from escaping.



Faraday's cage restricts the emission/interference within the defined area of a PCB.

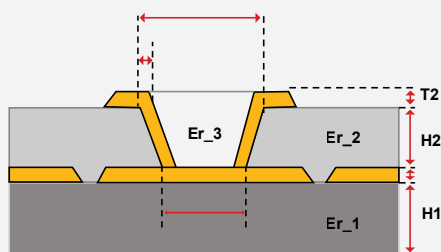
Via fence implementation

- Place ground floods or ground planes on multiple layers.
- Connect these planes with vias around the board perimeter.
- Space vias close enough to be effective at your frequencies.

Spacing rule

Via spacing should be less than $\lambda/20$ at your highest frequency of concern.

- If your highest problem frequency is 1 GHz (wavelength = **18 cm** in FR4 material).
- $\lambda/20 = 0.9$ cm
- Via fence spacing should be less than 9 mm.



PCB DESIGN TOOL

Via Impedance Calculator



TRY TOOL

5.5 Use filters to control conducted EMI

Filtering allows you to control electromagnetic fields associated with high-frequency noise. Filters help you:

- Keep noise away from sensitive components.
- Prevent noise from reaching structures that can radiate (cables, apertures).
- Block noise from entering your system through power or signal lines.

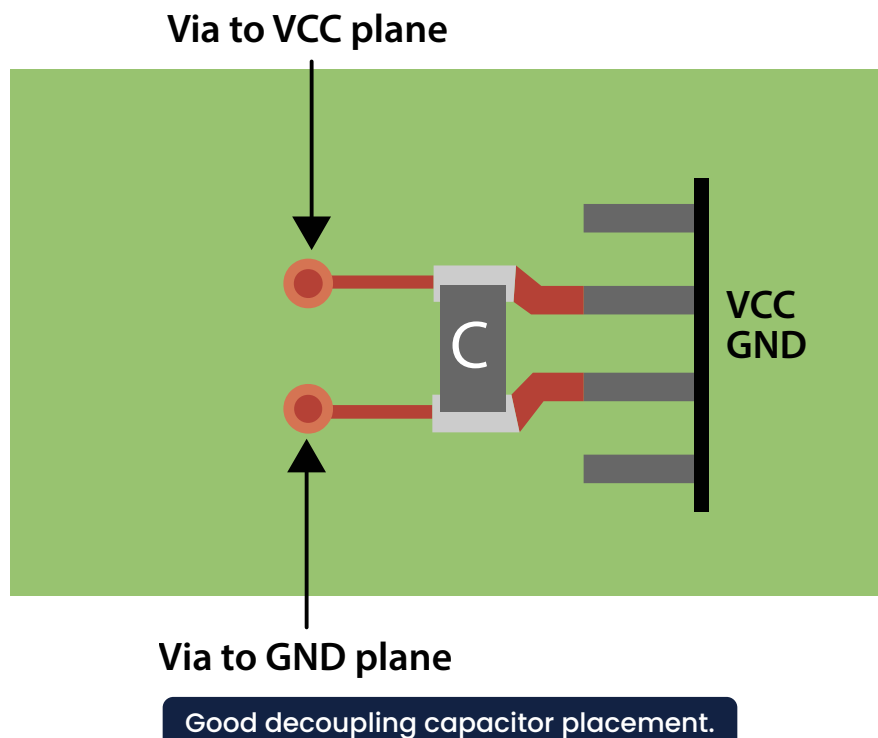
The following filtering methods help to prevent noise from coupling into cables and interfaces:

5.5.1 Place decoupling capacitors

When ICs operate, they switch current at high frequency. This transition creates noise on the power rails and traces connected to the IC.

How to use decoupling capacitors:

- Place decaps as close as possible to IC power pins (millimeters, not centimeters)
- Ground the capacitors directly using the adjacent ground plane
- Keep the loop area small (from power pin → capacitor → ground → back to IC ground pin)



For more information, read [decoupling capacitor placement guidelines every PCB designer should know](#).

5.5.2 Include EMI filters

For noise that travels beyond individual ICs, such as on power lines or I/O interfaces, use dedicated EMI filters.

Filters allow desired signals to pass through while blocking (attenuating) unwanted signals.

Various EMI filter types are available to suppress interference across different frequency ranges. These can be:

- **Passive circuits:** Using only capacitors, inductors, and resistors.
- **Active circuits:** Including active components like op-amps.

Different filters offer different levels of suppression bandwidth.

Choosing the right filter

The optimal EMI filter choice depends on several factors:

- **Board space:** How much room is available?
- **Required attenuation:** How much suppression is needed?
- **Power handling:** How much current must the filter handle?
- **Frequency range:** What broad spectrum or narrow frequency you are targeting?
- **Cost:** What is the budget constraint?

Filter placement strategy

Strategic filter placement is essential for maximum effectiveness. Place filters where signals enter or leave the system. It may include:

- Power supply input lines
- Signal connectors (I/O ports)
- Communication interfaces

Why this works

Filtering signals at the entry point ensures:

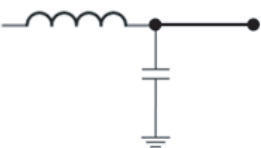
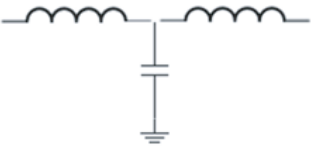
- External noise is blocked before entering the unit.
- Internal noise is blocked before exiting the unit.
- Cables don't carry high-frequency noise that can radiate.

Design rule

Filter before signals reach cables. They can become effective antennas for high-frequency noise.

Common passive filter configurations

There are several standard approaches for implementing low-pass filters using passive components. Selecting the right topology depends on the impedance of the source and the load.

Filter type	Circuit schematic
C	
L	
LC	
T	
Pi	

Basic passive filter types. Image credit: EMC United

When filters aren't enough

If you've implemented EMI filters and followed best layout practices but still have strong radiated EMI, you may need more aggressive measures. In these cases, consider adding shielding at the board or enclosure level, as discussed in section 5.6.

Ferrites

Ferrites offer a simple and effective way to suppress high-frequency noise on cables, power lines, and traces, especially when filters alone are not practical.

They add impedance to nullify high-frequency noise on traces, wires, or cables.

Ferrites work by:

- Increasing the inductance of a conductor, thereby increasing impedance at high frequencies.
- Allowing DC and low-frequency signals to pass with little effect.

Before using ferrites, consider these three factors:

1. DC and low-frequency saturation

Problem:

- Ferrites can saturate when exposed to DC current or strong low-frequency AC (50/60 Hz).
- Once saturated, they become much less effective at suppressing high-frequency noise.

Solution:

- Check the ferrite's saturation current rating.
- Choose ferrites designed for the current levels in your application.
- Consider alternative filtering for high-current applications.

2. Signal integrity effects

Problem:

- Ferrites add inductance to signal lines.
- This inductance can distort signal quality.
- Fast edges can be slowed or rounded.

When this matters:

- High-speed digital signals.
- Communication interfaces with strict timing requirements.
- Signals requiring sharp edges.

Solution:

- Verify that the added inductance doesn't violate signal integrity requirements.
- Test signals with ferrite in place.
- Consider lower-inductance ferrites or alternative filtering.

3. Frequency-specific materials

Ferrites are made from different materials. Each material blend is formulated for specific frequency ranges.

Design rule

Make sure the ferrite material targets the appropriate frequency range for your system.

Example of wrong material choice

Make sure the ferrite material targets the appropriate frequency range for your system.

- Concerned frequency: 250 kHz, causing conducted emissions.
- Ferrite chosen: High-frequency nickel-zinc material (effective above 100 MHz).
- Result: No improvement because the ferrite is ineffective at 250 kHz.

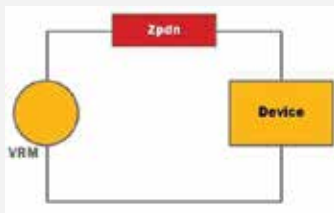
How to choose ferrites

- Identify the frequency range of your noise problem.
- Review ferrite datasheets for impedance vs. frequency curves.
- Select a ferrite with peak impedance in your problem frequency range.
- Verify effectiveness through testing.

Common ferrite forms

- **Ferrite beads:** Small components that slide over wires or mount on PCBs.
- **Ferrite sleeves/clamps:** Snap-on devices for cables (often seen on power cords).
- **Ferrite cores:** Used to wind multiple turns of wire for increased effectiveness.

Ferrites can be found in a variety of alternative forms, from ovals installed on ribbon cables to amorphous shapes customized to unusual installations.



PCB DESIGN TOOL

Power Distribution Network Analyzer



TRY TOOL

5.6 Implement shielding to confine EM fields

Shielding prevents electromagnetic fields from escaping or entering a system by forming conductive barriers that act as Faraday cages.

For shielding to be effective, it must be continuous, well-bonded, and properly terminated.

Poorly implemented shielding can behave like an antenna and make EMC performance worse instead of better.

Where shielding can be applied

Shielding can be implemented at three levels:

1. **PCB level:** Shields covering parts of a circuit board.
2. **Enclosure level:** Metal housing containing the entire system.
3. **Cable level:** Shielded cables connecting components.

If a unit has a non-conductive (plastic) enclosure, only PCB-level shielding can be truly effective.

5.6.1 PCB-level shielding

A PCB-level mechanical shield is a closed conductive container that covers all or part of a circuit board. The shield connects to the ground return plane.

It serves two main functions:

1. Protect circuits from external noise

- Keeps external EMI from affecting sensitive circuits.
- Protects receivers, low-noise amplifiers, and precision analog circuits.

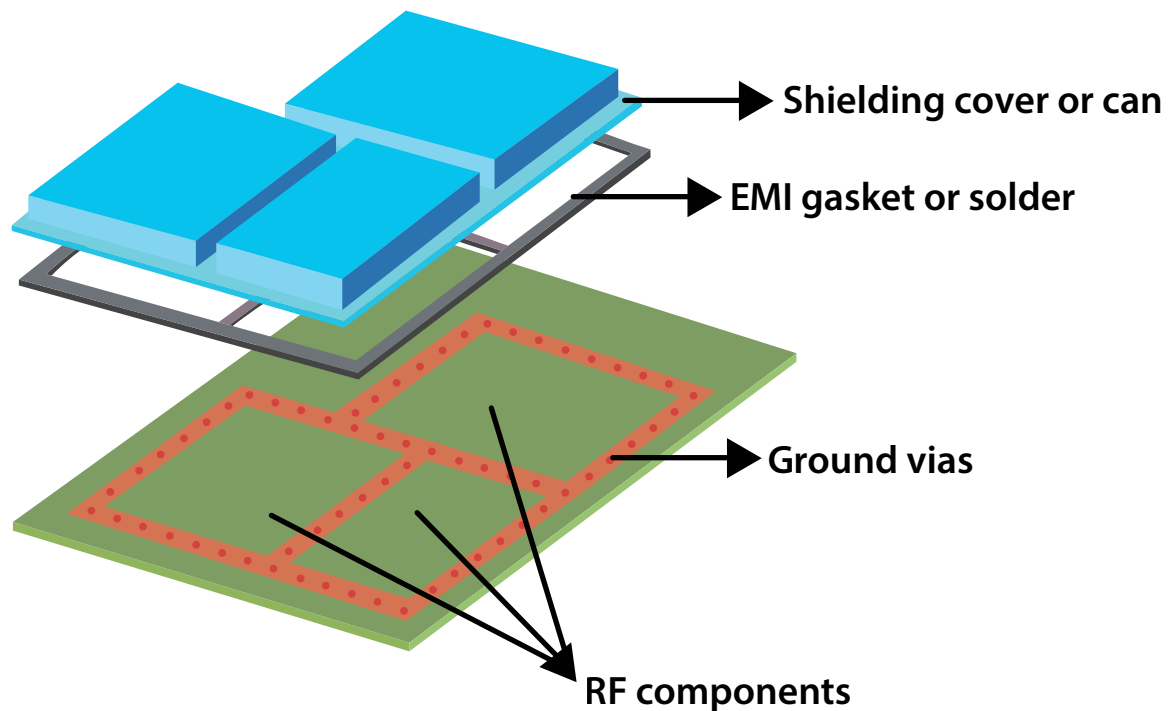
2. Prevent cross-coupling to other circuitry

- Contains noise from noisy circuits (oscillators, SMPS, processors).
- Prevents noise from spreading to other parts of the board.

What makes a good shield:

An effective shield should have features of a Faraday cage. They include:

- Continuous conductive enclosure on all sides.
- Multiple connection points to ground plane.
- Minimal gaps or openings.



EMI/EMC shielding protects the signal from external noise.

PCB shields are most commonly used when only a part of the system requires isolation or when no conductive enclosure is available.

Apertures in PCB shields

Real-world PCB shields often require openings for:

- Airflow and thermal management
- Component height clearance
- Access to adjustments or test points

Design rule to avoid slot antennas

As discussed in section 4.6, long slots can become effective antennas. To minimize this problem:

Prefer small circular holes over long slots

- Many small holes provide airflow without creating effective antennas.
- Keep hole diameter small relative to wavelength.
- Distribute holes evenly rather than creating large openings.

Rule of thumb

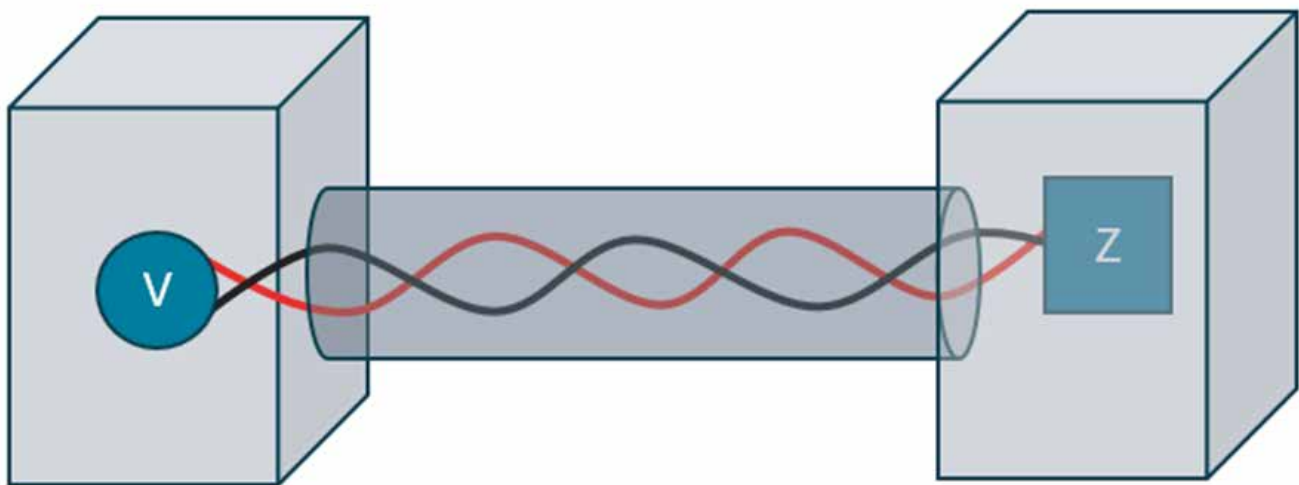
- Hole diameter should be less than $\lambda/20$ at your highest problem frequency
- Example: At 1 GHz (wavelength = 18 cm in FR4 material), keep holes smaller than 0.9 cm in diameter

5.6.2 Enclosure-level shielding

For stronger overall protection, move to enclosure-level shielding with a conductive housing.

A perfect Faraday cage provides complete electromagnetic shielding at the enclosure level. For electronics systems, this requires:

1. **Metal enclosure:** Conductive housing on all sides.
2. **360° cable shield termination:** Cable shields are connected completely around their circumference at each end.
3. **Termination to enclosure exterior:** Shields connect to the outside of the housing, not to internal PCBs.



Perfect Faraday cage arrangement with solid enclosures (no apertures) around the voltage source and the load, and a continuous solid shield around the connecting cable, bonded to enclosures at both ends. Image credit: EMC United

Why 360° termination matters

- Provides the lowest impedance connection.
- Prevents field leakage around the cable entry point.
- Maintains shield effectiveness at high frequencies.
- Prevents the shield itself from becoming an antenna.

Any cable passing through a conductive enclosure must be either:

1. **Shielded:** With shield terminated to the exterior of the housing.
2. **Filtered:** With a filter as close to the I/O point as possible.

Without these precautions, cables can carry high-frequency noise into or out of the enclosure, defeating the purpose of the shielding.

Connecting PCBs to enclosures

Do you need a PCB-to-enclosure connection?

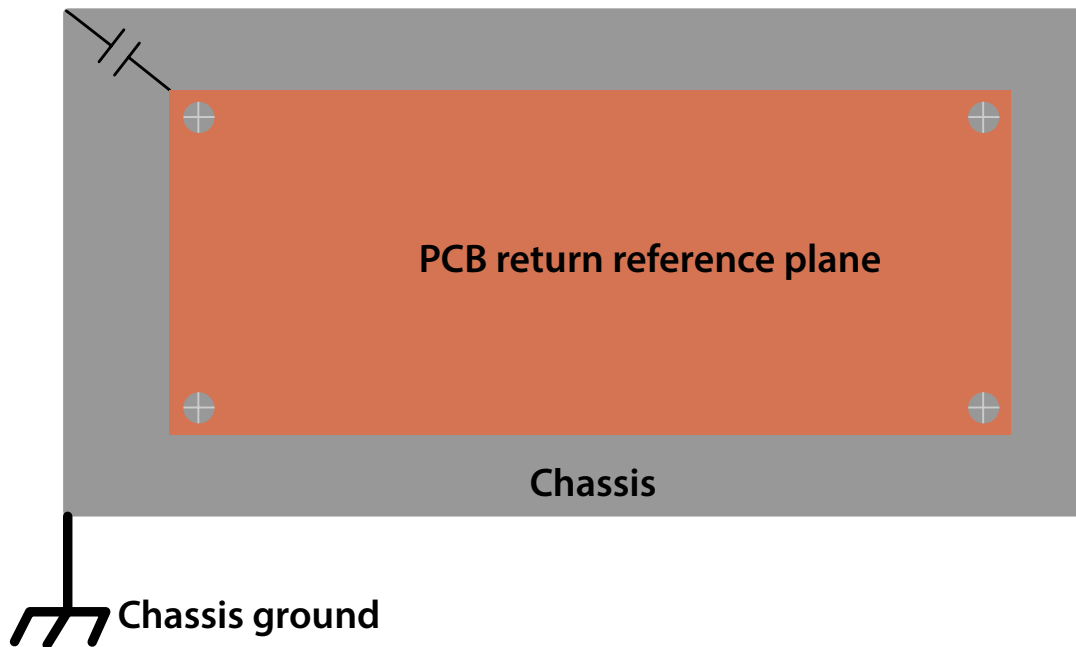
In a perfect Faraday cage configuration, the PCB would not need any direct connection to the housing. The enclosure would perfectly contain all fields.

Reality: Shields are never perfect

Since no shield achieves perfect containment, a connection between the PCB and housing may be appropriate.

Common implementation:

Use a capacitor connection from the PCB to the chassis through one or more mounting holes.



Chassis/PCB connection through a capacitor.

Why use a capacitor to connect a board to chassis

- Provides a low-impedance path at high frequencies (where EMI problems occur)
- Blocks DC and low frequencies (prevents ground loops)

5.6.3 Cable-level shielding

Cable shields are effective when:

- Shields terminate to the enclosure/chassis at both ends (not to PCBs).
- Termination is 360° around the cable circumference.

When cable shielding doesn't work

They are often ineffective or counterproductive when:

- Pigtail, drain wire, or single-wire terminations are used.
- Only one end is terminated, or both ends are left floating.
- Shields connect to PCB ground instead of chassis.
- Enclosures are plastic (non-conductive).

Why pigtails fail

A pigtail connection:

- Has high inductance, leading to high impedance at high frequencies.
- Allows field leakage around the connection point.
- Can cause the shield to act as an antenna.

Terminating a shield at only one end is the right approach only in certain limited situations. These usually involve:

- *Audio circuits*
 - *Fixed installations*
 - *Direct-to-board shield connections*
-

Connecting shields to PCBs

When units have conductive enclosures, cable shields should:

- Connect to enclosure or chassis.
- Not connected directly to PCBs.

Why?

The shield must be a part of the Faraday cage formed by the enclosure. Connecting to the PCB breaks this protection and can create ground loops.

Cable shielding with plastic enclosures

When PCBs are housed in plastic, cable shielding provides little benefit because:

- No metal enclosure exists to complete the Faraday cage.
- Shield has no chassis reference to connect to.
- Fields can escape through the plastic housing.

Better approaches for plastic enclosures

1. **Filtering:** Filter all signals at cable entry points.
2. **PCB-level shielding:** Shield critical circuits on the board.
3. **Cable management:** Route cables carefully and keep them short.

Special case: Shielded cables with plastic enclosures

If shielded cables are unavoidable in a plastic enclosure design (as when a shielded USB cable is used with unshielded electronics):

1. Connect through a common-mode filter

- Filter the shield connection to the PCB.
- Prevent the shield from conducting high-frequency noise on to the board.

2. Add transient protection if needed

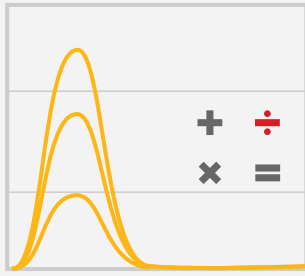
- If ESD is a risk.
- Protect sensitive circuits from voltage spikes on the shield.

Why these precautions help

- The filter prevents the shield from becoming a noise conductor.
- Transient protection prevents damage from static discharge.
- Precautions provide some benefit even without ideal shield termination.

Plastic vs. metal enclosure cost tradeoff

Metal enclosures cost more than plastic but provide inherent EMI shielding. When EMC is critical, a well-bonded metal enclosure can reduce overall project cost by avoiding expensive board-level shielding, extensive filtering, and multiple EMC test iterations. Evaluate total project cost, not just the enclosure cost.



PCB DESIGN TOOL

RLC Resonant Frequency and Impedance Calculator



TRY TOOL

Conclusion

EMC depends on proper system design

An electronic system is made of many components working together. If these components are not arranged properly, various EMC problems can occur.

Consider EMC for every component

When designing a PCB, think about how each component affects EMC. Ask yourself:

- Does this component generate high-frequency noise?
- Is this component sensitive to EMI?
- How can I route this signal (and noise) to contain its fields?
- Where should I place filters or shields?

Two paths to good EMC performance

Good EMC can only be achieved through proper PCB design practices. You must:

1. **Eliminate interference sources:** Prevent noise from being generated or radiated.
2. **Protect circuits from interference:** Shield sensitive circuits from external fields.

The goal is to maintain the intended functionality of the circuit board and pass EMC testing on the first attempt.

Electromagnetic noise is never welcome in a PCB design. However, it is inevitable. The key is controlling and managing it.

At Sierra Circuits, we take intensive care to ensure signals do not interfere with each other when it comes to traces, vias, and even PCBs operating in unison. EMC improvements with accurate board design do not add extra cost to the final product, so they are recommended during the initial phase.

EMC problems are expensive to fix late

Addressing EMC problems after manufacturing is costly. Design for EMC from the start of your project to avoid:

- Failed EMC testing
- Expensive redesigns and respins
- Project delays
- Budget overruns

Following the guidelines in this document will significantly increase your chances of passing EMC tests on the first attempt.

Helpful relationships

Decibels

Many parameters commonly used in EMC are expressed in decibel ratios. This simplifies a lot of mathematics and captures critical information about the relationships between different quantities.

Decibels are fundamentally logarithmic. To express a voltage in dB:

$$V_{dB} = 20 \log \frac{X (V)}{1 V}$$

That gives you the dBV (decibels compared to 1V) of a given voltage value. Similarly for currents you can write:

$$A_{dB} = 20 \log \frac{X (A)}{1 A}$$

To get dBA (decibels compared to 1 A). For power you can write:

$$W_{dB} = 10 \log \frac{X (W)}{1 W}$$

For dBW (decibels compared to 1W)

Power is 10 log instead of 20 log because of the square term in $P = V^2/R$

The most common decibel units are:

dB of Gain = the gain compared to the power of the original signal

dBW = power compared to 1 Watt

dBm = power compared to 1 milliwatt

dBV = voltage compared to 1V

dB μ V/m = electric field strength compared to 1 microvolt/meter

dB μ A/m = magnetic field strength compared to 1 microamp/meter

Decibels are always relative to some known quantity. They are always a ratio. There's no such thing as an electric field that is 20 dB strong. 20 dB compared to what? If a field is stated to be 6 dB over a limit, make sure you understand what the comparison unit is.

Useful properties of dB:

- In electric or magnetic field strength, voltage, or current, going up or down by 6 dB either doubles or halves a quantity.
- In power, going up or down by 3 dB doubles or halves the quantity.
 - If discussing power, then $P_2 = P_1/2$ expressed in decibels is $P_2 = P_1 - 3$.
 - If discussing voltage, then $V_2 = V_1/2$ expressed in decibels is $V_2 = V_1 - 6$.
- Multiplying voltage or current by 10 is a 20 dB increase; dividing by 10 is a 20 dB decrease.
- Multiplying power by 10 is a 10 dB increase; dividing by 10 is a 10 dB decrease.
- Because adding or subtracting is often easier than multiplying or dividing, decibels have an advantage.

Conversions

To convert	Use
Volts to dBV	$\text{dBV} = 20 \log(V)$
Volts to dB μ V	$\text{dB}\mu\text{V} = 20 \log(V) + 120$
dBV to Volts	$V = 10^{(\text{dBV}/20)}$
dB μ V to Volts	$V = 10^{((\text{dB}\mu\text{V} - 120)/20)}$
dBV to dB μ V	$\text{dB}\mu\text{V} = \text{dBV} + 120$

To convert	Use
dBμV to dBV	$\text{dBV} = \text{dB}\mu\text{V} - 120$
Power: dBm to dBμV	$\text{dB}\mu\text{V} = \text{dBm} + 107$ (given a $50\ \Omega$ system)

Log identities

- If $Y = \log(X)$, then $X = 10^Y$
- $\log 1 = 0$
- For numbers greater than 1, log is positive
- For numbers less than 1, log is negative
- $\log (A*B) = \log A + \log B$
- $\log (A/B) = \log A - \log B$
- $\log (A^n) = n*\log A$
- $\log 0$ is undefined

Frequencies

Frequency vs. wavelength in free space (air/vacuum)

Frequency	Wavelength (λ)	Quarter Wavelength ($\lambda/4$)
3 MHz	100 m	25 m
30 MHz	10 m	2.5 m
300 MHz	1 m	25 cm
1 GHz	30 cm	7.5 cm
2 GHz	15 cm	3.75 cm
4 GHz	7.5 cm	1.9 cm
8 GHz	3.75 cm	9 mm

The relationship between frequency and wavelength is dependent on the material through which the electromagnetic field is propagating. The wavelengths of a given frequency are different in FR4 than they would be in air. So, the dimensions of a structure that may make a good radiator on a PCB may be different than those that make a good radiator in free space.

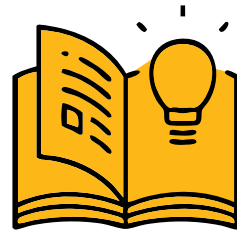
The following table assumes that the speed of light in FR4 is 0.6 of the speed of light in a vacuum, but depending on the precise material formulation that value can range from 0.4 to 0.8.

Frequency	Wavelength (λ)	Quarter Wavelength ($\lambda/4$)
10 MHz	18 m	4.5 m
25 MHz	7.2 m	1.8 m
50 MHz	3.6 m	90 cm
100 MHz	1.8 m	45 cm
200 MHz	90 cm	22.5 cm
500 MHz	36 cm	9 cm
1 GHz	18 cm	4.5 cm

Acronyms

AC	Alternating current
DC	Direct current
EMC	Electromagnetic compatibility
EMI	Electromagnetic interference
EMR	Electromagnetic resilience
ESD	Electrostatic discharge
IC	Integrated circuit
I/O	Input/Output
PCB	Printed circuit board
PWM	Pulse-width modulation
SMPS	Switched-mode power supply

Other resources on EMI and EMC



Articles:

- 7 Tips and PCB Design Guidelines for EMI and EMC
- Best EMI and RF Shielding Methods for Flex PCBs
- How to Reduce EMI to Improve Signal Integrity
- How to Mitigate EMI Issues in Your PCB Designs by Daniel Beeker
- How Grounding Controls Noise and EMI in a PCB
- Rick Hartley's PCB Design Recommendations to Minimize EMI

Case study:

- Case Study: Resolving EMI and PCB Routing Issues in a Medical Optical Scanner

On-demand webinars presented by industry experts

- 7 Habits to Successfully Pass EMC
- PCB Layout Guidelines and Grounding Techniques to Avoid EMI and Crosstalk
- From Theory to Practice: Implementing EMC Design Rules
- Unintentional Antennas: Why PCBs and Cables Start Radiating and How You Can Stop It
- PCB Design for Low EMI
- How Good is My Shield? An Introduction to Transfer Impedance and Shielding Effectiveness
- Designing for EMC and Preparing a Test Campaign

6. Sierra Circuits' manufacturing and assembly capabilities

Sierra Circuits has been serving PCB designers and engineers with the latest technologies since 1986 and has worked with over 20,000 customers since then. We specialize in manufacturing, assembly, and HDI technology. We handle all aspects of circuit board production.

We provide our customers with DFM and DFA support, which helps produce unprecedented board quality with high reliability. By choosing us, you can eliminate miscommunication between multiple vendors and delays since we provide a single point of support.

Sierra Circuits helps PCB designers plan it right! Our engineering staff has been trained on controlled impedance and can analyze the design from a holistic point of view. Our engineering support and our stack-up team provide valuable suggestions with their knowledge of controlled impedance for high-speed designs, analog/digital, high-density board manufacturing design rules, and design for assembly guidelines. Services include system-level design, schematic capture, circuit board layout, and PCB/PCBA DFM.



Our facility

Why Sierra Circuits?

Customer-oriented approach: We collaborate closely with you and tailor solutions to meet your unique design requirements.

Rapid high-tech innovation: We work in a fast-paced environment and implement cutting-edge technologies to bring your design ideas to life.

Integrated systems: Integrated turnkey world-class team that includes design, engineering, manufacturing, support services, and supply chain that engages at the beginning of all of our partnerships.

Focus ready: We focus on complex products for defense and aerospace that require high reliability and repeatability to support mission-critical systems on time.

Sierra Circuits' PCB manufacturing capabilities overview

Rigid PCBs

PCB highlights	Fabrication highlights	Quality management systems
Up to 30 layers	Routed arrays	AS9100D
Board thickness: .005" - .250"	Edge plating	ISO 9001:2015
Max panel size: 21" x 29"	Bevels	ISO 13485:2016
Min trace and space: .002"	Heat sinks	
Solder mask feature tolerance: .001"		
Blind and buried vias		

HDI and microelectronics

High tech in small form	Features	Certified for your industry
Special stack-up help / NPI review	Down to 1.5-mil trace space, 2-mil holes	Mil-spec
Up to 30 layers	Blind vias, buried vias, and other microvia techniques	Automotive
Files evaluated by engineers	Up to 5 sequential lamination cycles	Aerospace and defense
Quote sent by email within a few hours	Laser direct imaging	Medical devices
	via-in-pad technology	

Flex and rigid-flex PCBs

Surface finishes	Standard flex materials	Fabrication highlights	PCB classification
HASL	0.5 mil to 5 mil thick polyimide material	Routed array	IPC 6012, Class 1, 2, and 3
Lead-free HASL		V score	ISO:9001:2015
OSP (Shikoku F2 and Entek)	1 mil to 5 mil thick copper clad base material	Edge plating	ISO 13485: 2015
ENIG	Flame retardant laminate	Edge castellation	
Immersion silver	UL and RoHS compliant	Countersink	
Immersion white tin		Bevel	
Tin nickel		Milling	
Selective gold			

Aerospace and defense PCBs

Capabilities	Documents we provide	Ideal for
Layer count: Up to 30	Certificate of conformance	DOD product designers developing systems for mission critical functions
Minimum board thickness: 10 mil	Material specifications	
Thickness tolerance: 10%	Reflow profile copy (included with first article)	Contract manufacturers who need fast, predictable lead times
Minimum trace and space width: 3 mil	Photo requirements	
Minimum drill-to-copper clearance: 6 mil	First article inspection report	Purchasing teams who require pricing and supply chain transparency
Maximum panel size: 21" x 29"	Record of calibrated tools used during manufacturing	
Controlled impedance	FPT, AOI, ionic cleanliness report	Aerospace engineers looking to shore-up their supply chain troubles
Stacked vias		

Capabilities	Certifications
Operating frequency: 10 MHz to 30 GHz	PC Manufacturers Qualification Profile (MQP)
Sodium etch hole treatment process	IPC-A-600
Full-body gold plating up to 120u"	IPC-6012 standards
Soft gold without nickel or ENEPIG surface finishes for enhanced board protection	ISO 9001:2015
	NADCAP
	ITAR registered
Design assistance	RoHS/REACH standards

Sierra Circuits’ PCB assembly services

We offer a diverse range of PCB assembly services tailored to meet your unique requirements. Whether you need a full turnkey assembly, consigned or partially consigned assembly, we've got you covered.

Full turnkey PCB	Partially consigned assembly	Consigned assembly	Assembly capabilities
Design review	Some parts provided by the designer	Components are provided by the PCB designer	BGA, micro-BGA, QFN,CSP, lead less devices up to 0.35mm pitch, press fit components
Bare board fabrication	We will source the rest	Sierra Circuits will fully assemble your PCBs and ship them to you.	DFA
Component procurement	Supply chain resilient	Quick turn times	RoHS, leaded, indium, clean and no clean chemistry
Testing	Quick turnaround time	Fully transparent pricing	Component sizes: 0201, 01005, 08004
Quick turnaround time			Paste in-hole

To help you overcome the challenges involved in component procurement, Sierra Circuits has introduced customer-owned inventory service (COIN). It's a personalized component sourcing and stocking program. We procure and store parts for your PCBAs. This can help you significantly reduce your product development time.

Sierra Circuits' PCB design services

Whether you're beginning with a new design or enhancing an existing one, our design experts provide the assistance you need. Our PCB design services include stack-up design, signal integrity analysis, circuit board layout optimization, schematic capture, and reverse engineering.

Our capabilities include:

- High-speed digital designs, analog designs, mixed designs, power designs, and RF designs
- Controlled impedance with a tolerance of $\pm 5\%$
- Mechanical-constrained boards and enclosures
- Flex, rigid, and rigid-flex designs
- PCB designs for mobile phones
- HDI designs with via-in-pad, blind via, buried via, microvias, via stacking, and staggering technology
- Fine-pitch BGAs of 0.4 mm and 0.5 mm
- Designs with high-pin count full matrix BGAs 0.4 mm pitch, PoP package, with 2 mil/ 2 mil tracks/spacings
- Design for RoHS compliance
- Optimum impedance calculation for single-ended, differential, and coplanar-waveguide models

Advanced PCB design tools

From material selection to impedance calculation, streamline your design process by using our comprehensive suite of cutting-edge PCB design tools. These web applications enable you to expedite your design process and make the right decisions to ensure signal integrity and reliability in your circuit board.

Sierra Circuits PCB certification and registration

IPC certified for quality standards in PCB fabrication and assembly

- IPC Manufacturers Qualification Profile (MQP) for Sierra Circuits, Sunnyvale, California
- IPC-A-600 inspectors

ISO certified for military, aerospace, and medical device PCB fabrication and assembly

- ISO 9001:2015 certificate
- AS9100D - military and aerospace
- ISO 13485:2016 certificate - medical devices

Mil-Spec certified and DLA approved

- Acc 2020 CVI VQE 035113
- JCP certification approval letter

ITAR Registered

- ITAR registration letter

Underwriters Laboratory certified

- Flex and rigid flex PCBs
- Rigid PCBs

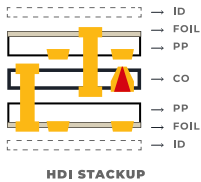
Exceptional material compliance and reporting

- RoHS / REACH compliance material composition declaration
- Conflict minerals reporting template

Certified minority supplier by NMSSDC

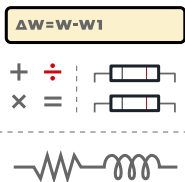
- Minority business certification

The Designer's Toolkit

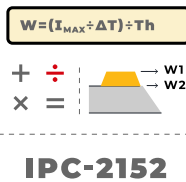


HDI STACKUP

Stackup Designer

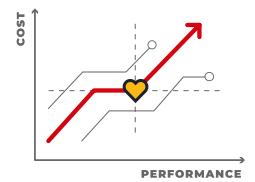


Impedance Calculator

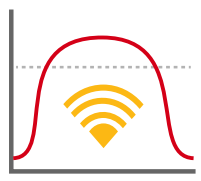


IPC-2152

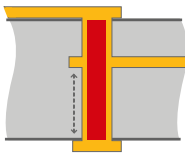
Trace Width,
Current Capacity and
Temperature Rise
Calculator



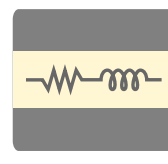
Material Selector



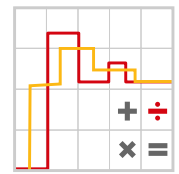
Bandwidth, Rise Time
and Critical Length
Calculator



Maximum Via
Stub Length
Calculator



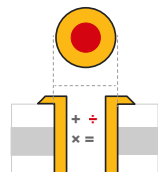
Signal Layer Estimator



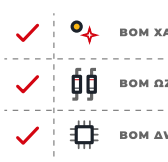
Transmission Line
Reflection Calculator



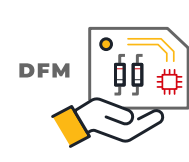
RLC Resonant
Frequency &
Impedance Calculator



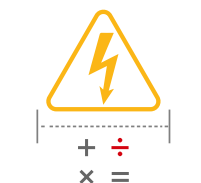
Via Current
Capacity and
Temperature Rise
Calculator



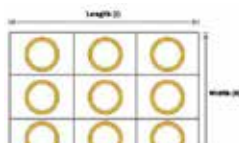
BOM Checker



Better DFM



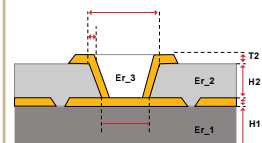
Conductor Spacing
and Voltage Calculator



Via Thermal
Resistance Calculator



Power Distribution
Network Analyzer



Via Impedance
Calculator

Try Tools



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